Social Learning and Sentiment Contagion in the Bitcoin Market*

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Abstract

Using novel data on investors' social interactions, we document sentiment contagion as a direct consequence of social learning in the Bitcoin market. Our findings suggest that the social learning process is inefficient, as investors respond positively to social sentiment, but social sentiment does not positively predict returns. Echo chamber effect and selective interpretation of information may simultaneously contribute to such inefficiency. We further confirm that sentiment contagion is not a sideshow: at the individual level, sentiment contagion predicts the direction of trading conditional on its occurrence; at the Bitcoin market level, the aggregated sentiment contagion index positively predicts market trading volume and volatility. Moreover, we highlight the error-prone nature of social learning, wherein the socially constructed optimism predicts Bitcoin crashes. Finally, we establish a link between social learning and bubbles: the elevated propagation of optimism is highly correlated with the trading volume.

Keywords: Social Finance, Sentiment Contagion, Bubbles, Bitcoin

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*The views expressed in this paper are those of the authors and do not necessarily reflect the position of the Federal Reserve Bank of Dallas, or the Federal Reserve System.

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1 Introduction

"*Man is by nature a social animal*"- Aristotle

Like Aristotle, researchers have long realized the importance of social activities on economic outcomes. However, the role of social interactions has been largely absent in economics and finance until recent decades (Shiller et al. (1984); Shiller and Pound (1989); Shiller (2007); Hirshleifer (2020)). In his pioneering works, Robert Shiller argues that social learning, among other factors, plays a pivotal role in influencing investors' decisions as they update their beliefs through social interactions. Shiller further proposes a potential link between social learning and asset pricing dynamics. Despite its prominence in the literature, direct empirical evidence on social learning has been limited, possibly due to data constraints.

We provide novel insights into social learning by leveraging new data on investors' social activities in the Bitcoin market. We document direct evidence for social learning and shed light on whether social learning is efficient. We further document the aggregate impact of social learning on market outcomes. The Bitcoin market is well-suited for our study due to its distinctive social characteristics. First, Bitcoin, as a product born in the internet era, has its origins and development closely tied to social networks. Right after the inception of Bitcoin, its founder, Satoshi Nakamoto, quickly established Bitcointalk, an online social platform designed to support Bitcoin operations and facilitate information sharing and communications among Bitcoin users. Over time, Bitcointalk has grown into the most influential social platform on Bitcoin. Second, as a novel asset class, investors have limited understanding of Bitcoin, making its valuation relatively difficult to determine. Users often turn to social platforms to share information and views. Third, as suggested by recent studies (Cong et al. (2021)), the valuation of Bitcoin depends on the aggregate demand from the social network, which further relies on social adoption. Rational investors would engage in social interactions to gather information on such social demands. Moreover, mistaken beliefs can form and spread via social interactions among investors (e.g., Hirshleifer (2020), Han et al. (2022)). Kogan et al. (2023) conjecture that retail investors hold a model of cryptocurrency prices, where positive returns increase the likelihood of future widespread adoption, which in turn will drive up asset prices. In fact, the Bitcoin market exhibits high volatility and has been linked anecdotally to bubble-like characteristics. This correspondence is unlikely coincidental, and instead consistent with the long-standing hypothesis on the role of social dynamics in

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fueling bubbles (Shiller et al. (1984), Burnside et al. (2016)).

We utilize Bitcointalk data in our study of social learning. Given its influence on Bitcoin investors, we view social learning on this forum as indicative of broader social learning trends within the Bitcoin market. Moreover, employing this social platform offers several other advantages. Firstly, Bitcointalk enables investors to share opinions and engage with one another through posting and replying to each other's posts. As a result, we are able to observe conversations among investors, which is one of the most direct forms of social interactions. By leveraging Natural Language Processing (NLP) techniques, we can determine the sentiment of each message in these conversations. Consequently, we can examine the overall sentiment of the conversation as well as the sentiment of investors before and after each interaction. This setting provides an ideal environment for studying social learning. Furthermore, a subset of Bitcoin investors on this platform voluntarily disclose their Bitcoin addresses to enhance the security of their forum accounts. By linking the sentiment expressed in their posts to their transactions, we can draw unique conclusions about the impact of social learning on investor trading behavior and market outcomes.

We begin by examining whether and how investors learn in social activities. The basic unit of observation in our analysis is a conversation segment which starts and ends with two posts (post[0] and post[1]) from the same investor. We use the term *social sentiment* to refer to the average sentiment of other investors' messages between post[0] and post[1]. The investors' social learning within the conversation is manifested by the way their sentiment changes in response to social sentiment. We find that investors' sentiment updates in the same direction as the social sentiment, suggesting that sentiment spreads via conversations on the social network. Specifically, investors become more bullish (bearish) after being exposed to positive (negative) sentiment by other posters. Our results are robust to controlling for known factors related to belief changes as well as time- and user-fixed effects. Therefore, our results indicate that social learning leads to sentiment contagion.

We then examine how users with different characteristics respond to social sentiment. Consistent with our expectation that sophisticated and informed investors are affected by biased social learning, we find that investors who are less sophisticated, less socially connected, and less informed are more susceptible to social sentiment. For example, inexperienced users exhibit, on average, an additional sentiment adjustment of 3.585% after engaging in a conversation with a social sentiment score of 1. Less informed users, compared to informed users, tend to update their sentiment by an additional 5.137% after participating in a conversation

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with a social sentiment score of 1. Furthermore, we consider users within a network structure and treat two users as connected if they have participated in at least one common conversation. We find that investors who have more connections are less responsive to social sentiment.

Our research uncovers inefficiencies in the process of social learning. On the one hand, sentiment in the posts on Bitcointalk does not significantly predict Bitcoin returns in the near term. In fact, as the forecasting horizon extends, social sentiment even exhibits a negative predictive power. On the other hand, investor beliefs are strongly influenced by the conversations as if they treat the social signals as informative. Furthermore, sentiment contagion is unrelated to the amount of news coverage as reported by RavenPack database. But it intensifies on days with greater Bitcoin volatility or higher dispersion in the RavenPack Bitcoin news sentiment, suggesting a greater reliance on social learning during periods of high uncertainty.

To gain a deeper understanding of such inefficiency, we highlight two patterns in social learning related to confirmation bias. Firstly, users selectively participate in conversations that are consistent with their priors, resulting in an echo chamber effect. For instance, users with positive prior sentiments are, on average, 6.734% more likely to engage in a thread initiated by a post with positive sentiment, compared to users with non-positive priors. To provide context, the unconditional probability of participating in a conversation that indicates a positive sentiment is 37.310%. Secondly, users selectively interpret information, responding more assertively to the social sentiment that aligns with their own prior sentiments.

Taking a step further, we examine the impact of sentiment contagion on investors' trading decisions. We link investors' sentiment changes following a conversation to their subsequent trading decisions within a short time window. We document that, unconditionally, investors' sentiment changes after a conversation do not predict individuals' future trading decisions. Such low sensitivity is mainly a result of low trading frequency: once conditional on trading occurs, sentiment changes significantly predict the direction of future transactions. If investors' sentiment improves by one unit, the conditional probability of actively buying Bitcoins during the conversations (resp. over the seven days after the conversations) increases by 9.244 (resp. 0.576) percent. This finding is robust with or without additional controls for news arrivals, market fluctuations, and overall sentiment on the Bitcointalk community. At a broad level, these results establish a micro-foundation for understanding the influence of social interactions on market outcomes.

We further document the aggregate impact of social learning on market dynamics. Specifically, we

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construct an index (SCI) to track the intensity of sentiment contagion in the market. We count the number of investors on Bitcointalk who change their sentiment in the same direction as the social sentiment in conversations and, after removing the time trend, use it as our proxy for the amount of social learning via conversations on Bitcointalk. As changes in investor sentiment are linked to individuals' future trading behaviors, we conjecture that SCI could also capture the aggregate demand for position changes in the future and, therefore, predict market dynamics such as trading volume. Consistent with our conjecture, we observe a strong predictive power of SCI for both Bitcoin trading volume and return volatility.

As sentiment spreads through the population via social interactions, more users become influenced by it. How do the accumulated belief changes via social interactions affect market outcomes? Given the documented inefficiencies in social interactions, we hypothesize that sentiment contagion may destabilize the market and contribute to the boom and bust in the Bitcoin market. We find confirmatory evidence for this hypothesis. First, we establish a link between social learning and bubbles. In our identified bubble episodes, we observe widespread optimism and a disproportionately high fraction of new investors engaging in these conversations. The contagion of optimism gets amplified in the bubble episodes and drives up the trading volume. We find that the correlation between the number of investors positively influenced by social interactions and the trading volume during bubbles can be as high as 0.669. Therefore, social learning offers a fresh perspective for comprehending the elevated trading volume during bubbles. Second, when the prevalence of optimistic views in the forum rises, the likelihood of a crash in Bitcoin returns substantially increases. This finding supports the error-prone nature of the impact of social interactions on aggregate outcomes proposed in David Hirshleifer's presidential address (Hirshleifer (2020)).

Our paper connects to several strands of literature. First, there is a growing interest in the social interaction literature on how investment ideas are transmitted. Shiller et al. (1984) argues that investment in assets is a social activity and Shiller and Pound (1989) considers the role of social interaction on the transmission of financial information. Han et al. (2022) offers a model of social interactions to understand the transmission pattern of investment ideas. Consistent with their spirit, our paper provides direct evidence for the effect of social interactions on sentiment contagion.

Our paper closely relates to an expanding body of literature that examines investors' behaviors on social networks. By leveraging data from an investor social network in the foreign exchange market, Simon and

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Heimer (2012) document that social interaction promotes active trading. Heimer (2016) further documents an increase in the level of the disposition effect after interacting with other investors. In the context of the investor social network StockTwits, Cookson et al. (2023) document that investors tend to selectively expose themselves to information aligning with their prior views, resulting in sustained disagreement. They also link this "echo chamber" effect to high trading volume in financial markets. In comparison, we highlight sentiment contagion as a direct consequence of social learning and document how social learning predicts individuals' trading decisions and aggregate market outcomes. Additionally, we utilize social learning to account for the significant trading volume during bubbles.

Our paper is related to studies that analyze the influence of peer actions. Hong et al. (2004) and Brown et al. (2008) provide evidence consistent with the notion that individuals are more likely to participate in the stock market when their geographically proximate peers participate. Hong et al. (2005) also shows that investors tend to buy stocks their local peers have been buying in the recent past. Compared to these studies, our data allows us to directly observe and measure social interactions and their impacts on market outcomes.

Do peer effects "wash out" when aggregated, or do they result in significant cyclical fluctuations in asset prices, as suggested by Robert Shiller? Despite the significance of this question, there have been relatively few empirical studies conducted. One notable exception is Kuchler et al. (2022), who examine the Social Connected Index based on friendship links on Facebook. Their findings reveal that investors show a greater inclination to invest in companies located in socially connected regions, and firms in regions with stronger social ties tend to have higher valuations and liquidity. In contrast, our study focuses on direct evidence of social interactions and documents sentiment contagion as a consequence of social learning among Bitcoin investors. We document several connections between social learning and market outcomes, including future market volume and volatility, Bitcoin crashes, and high trading volume during bubbles.

Our paper is very related to Huang et al. (2021). Using stock-financed M&A as an exogenous shock, they document the contagion of abnormal trading activity from "infected" investors to their neighboring investors. Their findings are consistent with the mechanism in which investors communicate with their local peers on trading decisions. They also estimate the rate of communication among different investors. Similar to their results, we also highlight the concept of contagion, but our emphasis is on change in investor sentiment or belief resulting from social interactions and the market impact of sentiment contagion. We document novel

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connections between sentiment contagion and the subsequent trading decisions at the individual level. We also link sentiment contagion to high trading volume during bubble episodes.

Most prior works on investor sentiment focus on constructing sentiment indices at the aggregate level to understand asset pricing dynamics. For example, Baker and Wurgler (2013) constructs a sentiment index using market variables such as IPO indicators and trading volume at the aggregate level to explain many asset pricing facts. We expand this literature by providing micro-foundations for sentiment dynamics, with an emphasis on the role of social interaction for propagating the spread of sentiment at the individual investor level.

The remainder of the paper is organized as follows: Section 2 describes the data and explains the main variables used in the analysis. Section 3 documents the sentiment contagion pattern. Section 4 discusses the inefficiencies in social learning and section 5 highlights patterns related to confirmation bias to understand such inefficiency. Section 6 presents the impact of sentiment contagion on individuals' trading decisions and on the aggregate market. Section 7 establishes a link between social learning and bubbles. Section 8 concludes.

2 Data

2.1 Bitcointalk

Our research study focuses on the Bitcoin market as it provides valuable opportunities to trace the full history of users' social interactive activities since its inception in 2009, in the era of the Internet. To gather data on social interactions, we primarily rely on Bitcointalk, an online forum that is considered to be one of the oldest forums in the Bitcoin space. What makes Bitcointalk special is that its founder, Satoshi Nakamoto, is the presumed pseudonymous author (or authors) of the original Bitcoin white paper, which describes Bitcoin's reference implementation. This endorsement by the founder of Bitcoin has made Bitcointalk one of the most active online platforms for Bitcoin users, with over 54 million messages posted for over 1.2 million topics and more than 2.7 million registered users as of March 2020. Bitcointalk also holds significant influence on Bitcoin trading and the real world, serving as a hub for traders to discuss market trends, share trading strategies, and provide insights into the direction of the market. It is frequently cited as a primary

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source of information and analysis in articles reporting on the latest developments in the Bitcoin space, including new projects, market trends, and regulatory issues. Its popularity has been widely covered by influential media outlets such as the Wall Street Journal, Forbes, and Bloomberg. Considering its impact on Bitcoin investors, we perceive social interactions on this forum as a proxy for the general trend of social activities in the Bitcoin market.

In this paper, our research is centered around posts on Bitcointalk, which are timestamped messages containing sentences written by users. Generally, sentences within each post are concise in nature. These posts are organized within threads, which are collections of sequential posts that revolve around a common theme, allowing users to interact with each other. As an example in our sample, Figure 1 presents a thread titled "What does the future of bitcoin look like" joined by user DavidLuziz on May 31, 2018, with subsequent participation from other users sharing their opinions.

On Bitcointalk, threads are displayed under different "child boards", which are subforums that gather discussions related to specific topics. For our study, we specifically focus on the "speculation" child board, where users predominantly discuss their views on Bitcoin's prospects. The "speculation" child board serves as an ideal field laboratory for our research, as we can examine the impact of social interaction on user sentiment, trading behavior, as well as aggregate Bitcoin prices and volumes.

There are multiple layers of user heterogeneity on this forum. For instance, we observe variations in the level of sophistication among users. Bitcointalk utilizes a merit system, where users are granted merits as points for their contributions to the Bitcoin community. Based on their merits and activities, users are assigned different ranks on the Bitcointalk website, ranging from high ranks such as "Legendary" to lower ranks like "Brand New". Typically, users with higher ranks are more sophisticated in their understanding of Bitcoin. Furthermore, a significant portion of users voluntarily disclose their age, gender, and home country, allowing us to study how sentiment contagion varies across different user characteristics.

2.2 User Sentiment

2.2.1 Textual Analysis and User Sentiment

In this paper, "user sentiment" refers to users' beliefs about future Bitcoin prices. We develop a methodology to extract user sentiment from posts collected on the Bitcointalk forum. Since our raw data

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consists of over 1 million posts, far beyond what we can manually interpret, we employ a textual analysis algorithm. Our algorithm is a two-step procedure based on a keyword dictionary and a natural language processing (NLP) algorithm developed by the Stanford University NLP group.

In the first step, we randomly select 10,000 sentences from our dataset and manually label them into four categories: "positive", "neutral", "negative", and "irrelevant". This approach mainly follows the methods used in Baker et al. (2016) and Tetlock (2007). We then use these labeled sentences as our training set and construct a keyword dictionary for each category. For example, keywords for the "positive" category may include "buy", "increase", and "rise", while keywords for the "negative" category may include "sell", "decrease", and "plunge", and so on. The dictionary for the "neutral" category includes terms such as "hold", "wait", "unpredictable", and others. If a sentence does not contain any of the keywords in the above three categories, we label it as "irrelevant". For our analysis, we focus on relevant sentences.

One challenge we face in our labeling exercise is distinguishing between descriptive statements about past performance and beliefs about the future in certain posts. For instance, a sentence like "Bitcoin market really increased a lot in the past months" can be ambiguous in terms of an user's opinion about future Bitcoin prices. To tackle this challenge, in the second step, we leverage the Stanford NLP algorithm to identify the tense of each sentence and detect backward-looking and forward-looking statements simultaneously. For example, if a sentence contains the keyword "increased", the Stanford NLP would detect the past tense, and our algorithm would label it as "irrelevant". Similarly, the Stanford NLP algorithm can identify future tense in sentences with phrases such as "will increase". Furthermore, our definition of forward-looking statements aligns with the format recommended by the SEC for 10-K reports, including sentences with terms like "expects", "anticipates", and so on.

After applying labels from the aforementioned two steps, we are able to extract user sentiment from each sentence. Additionally, we process sentences containing negative particles like "not" and "couldn't". If a sentence with negative particles contains words that are classified as "positive", we reverse our interpretation and label the sentence as "negative" instead of "positive". We assign a value of 1 to sentences with a "positive" sentiment, -1 for "negative", and 0 for all others. For example, a sentence like "Bitcoin price will roar" is assigned a value of 1, while "Bitcoin price is not going to fall" is also assigned a value of 1. Conversely, a sentence like "Bitcoin is doomed to fall" is assigned a value of -1, and a sentence like "Bitcoin

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price is really unpredictable" takes a value of 0. To calculate the overall sentiment of a post, we take the average of the sentiment values of all its sentences. As a result, the sentiment measurement for a post falls within a continuous range from -1 to 1. The out-of-sample accuracy of our algorithm is approximately 85%.

2.3 Transaction Data

We are able to link the sentiment of a subsample of users to their trading behaviors. This group of users voluntarily published their Bitcoin wallet addresses in a thread officially initiated by the forum organizer, as a protective measure for their Bitcointalk accounts against hacking.[1] Since Bitcoin transactions are publicly available on the Bitcoin Blockchain, we can trace their detailed transactions information such as trade size and trade timestamp using these published wallet addresses. Out of the approximately 37,262 users in our sample, we were able to link sentiment to trading decisions for 1284 users.

2.4 Other Data Sources

To systematically capture the impact of news events on the Bitcoin market, we obtain news related to Bitcoin from Ravenpack News Analytics, a reliable source that tracks news reports about Bitcoin and provides sentiment scores for them dating back to 2011. We refer to the general tone of news coverage on Bitcoin in the media as the news sentiment, and include it as a control variable in our analysis.

We gather Bitcoin market data, including returns and trading volume on an hourly basis from CoinAPI. We also calculate daily return volatility. This data is also utilized in a publication Griffin and Shams (2020).

[1]This is an official activity organized by the management team at the Bitcointalk forum. Users of Bitcointalk are recommended to disclose their public Bitcoin wallet address within this thread. When a user posts their public Bitcoin wallet address on Bitcointalk, they are essentially creating a public record of their ownership of that address. If their Bitcointalk account is later hacked, they can use their private key to create a digital signature that proves their ownership of the Bitcoin address they previously shared. This digital signature can then be used to authenticate the account holder's identity and facilitate the recovery of their Bitcointalk account. The digital signature is essentially a unique code that is created by applying a cryptographic algorithm to a specific set of data, which in this case is the user's Bitcoin address and a unique message or transaction. The signature can be verified by anyone using the public key associated with the user's Bitcoin address, which allows for the authentication of the user's ownership of the address without revealing the private key itself.

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2.5 Summary Statistics

Our main dataset of Bitcointalk posts covers the time period from May 1st, 2012, to July 30th, 2022. In Panels A, B, and C, we present summary statistics for user sentiment at the user, daily, and conversation levels, respectively. Panel D reports summary statistics for market information of Bitcoin, RavenPack news sentiment and Google Search Volume.

In Panel A of Table 1, we present the summary statistics for posting activities at the user level. For a representative user in our sample, the average sentiment score of the posts they published is 0.324, suggesting a general optimism towards Bitcoin among users. Such optimism is consistent with the existing literature on motivated beliefs (Bénabou and Tirole (2016)) and utility-based biases in beliefs (Brunnermeier and Parker (2005)). Furthermore, we observe a significant within-user standard deviation of sentiment at 0.608, indicating substantial variation in sentiment over time at the individual level. We require at least two posts for a given user to compute the standard deviation of post sentiment. This explains the smaller number of observations. The distribution of the number of posts is skewed, as evident from the difference between the median and mean number of posts per user.

Panel B of Table 1 describes post activities within a day. On a representative trading day, the average sentiment is 0.279, which is optimistic and consistent with the statistics at the user level. On a representative trading day, about 120 users participate in threads and publish around 178 posts.

Panel C reveals that posts within threads often exhibit dispersed sentiment, with an average within-thread standard deviation of 0.646. Furthermore, user engagement is notable, with 25 users participating in a representative thread that sees a total of 35 published posts. Overall, users seem to pay close attention to the posts, resulting in active exchanges within threads, with a median gap between consecutive posts within a thread of only 48 minutes.

Panel D provides summary statistics for variables at the aggregate market level. The annualized daily return is 119.1%, and the second row presents the annualized within-day volatility of Bitcoin, calculated using hourly return data. On average, the volatility is approximately 71.3%. When combined, the first and second row indicate a Sharpe ratio of approximately 1.670. We present volume-related variables for Bitcoin, spanning from the third to the fifth row. On a typical trading day, there are 16,476 transactions involving 9,361 bitcoins being traded. The average daily trading volume totals 56.514 million dollars. Moving to

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the sixth row, we present summary statistics for the sentiment of RavenPack News. In its original form, the sentiment score for RavenPack News Database ranges from 0 to 100, but we have normalized it to the range of $[-1, 1]$. On average, the transformed sentiment of news within a single day is 0.041 with a standard deviation of 0.330. The daily correlation between RavenPack News sentiment and post sentiment is 0.262 (not tabulated), indicating that while there is some overlap, these two data sources capture distinct dimensions of sentiment.

3 Social Learning and Sentiment Contagion

In this section, we investigate whether social learning propagates the spread of user sentiment, a pattern we refer to as *sentiment contagion*. Specifically, we examine how an user's sentiment may be influenced by the sentiment of others, as expressed through conversations on the forum.

3.1 Data Structure

Figure 1 illustrates an example conversation from the forum. User DavidLuziz published one post at 05:37:35 AM on May 31st, 2018 (post[0] at the top). Subsequently, several other users joined the conversation and shared their views about Bitcoin. At 07:50:10 AM on the same day, DavidLuziz published his second post in the same thread (post[1] at the bottom), becoming more optimistic about Bitcoin after interacting with other users.

We study consecutive pairs of posts from the same user, similar to the ones by DavidLuziz in Figure 1. We denote the first post published at timestamp t_0 as post[0], and the second post made at timestamp t_1 as post[1], as illustrated in Figure 2. We refer to the sentiment in post[0] and post[1] as the *prior sentiment* and the *ex-post sentiment*, respectively.

Between a consecutive pair of posts by a user, other users may publish posts in the same threads. We formally define these additional posts as a *conversation*. To measure the views of others in their interactions, we calculate a measure called *social sentiment* as the average sentiment of posts within the conversation. It's worth mentioning that an user's consecutive posts could be in two different threads.[2] In such cases,

[2]After posting in one thread, the user will receive notifications from the forum if others publish follow-up posts in the same thread. Therefore, users will pay attention to subsequent posts in the thread to which post[0] belongs.

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conversations refer to posts published between t_0 and t_1 in both threads to which post[0] and post[1] belong. Our results remain robust to restricting to a smaller sample with post[0] and post[1] in the same thread.[3]

To understand how sentiment spreads during social interactions, we examine how social sentiment influences the user's sentiment change, defined as the revision from prior sentiment to ex-post sentiment. We control for simultaneous confounding factors that might affect sentiment change, such as Bitcoin return and volatility, activities in other conversations within the forum and news arrivals documented in the RavenPack database.[4]

3.2 Time Window for Consecutive Posts

The time window allowed between consecutive pairs of posts by the same user in our analysis involves a trade-off. A shorter time window would reduce the effects of unobservable factors on sentiment change, but it could also result in a small sample, reducing the statistical power of our analysis. Our goal is to strike the right balance by choosing a window that is short enough to capture relevant changes but does not lead to excessive data loss. Figure 3 presents the cumulative distribution function of the time gap between consecutive pairs of posts by the same user in our sample. Over 50% of the pairs are within a 24-hour window. Hence, we choose a 24-hour time window for our main analysis. We also conduct robustness tests on the time window and find that a 24-hour window allows us to report a relatively conservative effect of sentiment learning.[5]

3.3 Evidence for Social Learning

To examine the effect of social learning, our main regression model is:

$$\begin{aligned} \text{Senti Change}_{i,j,t_0\to t_1} &= \beta_1 \text{Social Sentiment}_{i,j,t_0\to t_1} \\ &+ \gamma' \text{Control}_{i,t_0\to t_1} + \text{Fixed Effects} + u_{i,t_1}, \end{aligned} \tag{1}$$

where $\text{Senti Change}_{i,j,t_0\to t_1}$ is the revision in sentiment from post[0] to post[1] by user i in conversation j. As mentioned above, the timestamps for post[0] and post[1] are denoted as t_0 and t_1, respectively.

[3] See Appendix Table A4.

[4] Timeline in our analysis is illustrated in Figure 2.

[5] In Appendix Table A5, we show that social learning is stronger for longer windows.

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Social Sentiment$_{i,j,t_0\to t_1}$ is the social sentiment that user i has encountered in conversation j between time t_0 and t_1. Control$_{i,t_0\to t_1}$ is a set of control variables, including Bitcoin return and volatility, activities in other conversations within the forum and news arrivals documented in the RavenPack database. We also include lagged news arrivals as controls to account for response delays. Additionally, we control various sets of fixed effects such as user and day fixed effects. Standard errors are clustered at user and day levels.

One set of fixed effected included in our regressions is for the signs of the prior sentiment. This is motivated by the fact for a user with a positive sentiment, the room for being more optimistic is naturally smaller than someone with a negative sentiment.[6] Appendix Table A1 verifies that our findings are robust to using the ex-post sentiment as the outcome variable. We use sentiment change as the main dependent variable because we view it as a more direct measure to capture whether users respond to social learning. Appendix Table A1 finds that the magnitudes become larger when we use ex post sentiment as the outcome variable than having sentiment change on the left hand side.

The social learning channel predicts a positive coefficient for social sentiment as evidence that forum users revise their sentiments in the direction of sentiments in the conversations, Table 2 confirms this hypothesis.[7] Column 1 studies the relationship between sentiment change and social sentiment controlling for the directions of prior sentiment. A one standard deviation increase in social sentiment is associated with an increase in changes in sentiment of 3.752% (calculated as $42.177\% \times 8.896\%$). In columns 2 and 3, we further add user and day fixed effects, respectively to control for user characteristics[8] and day-specific confounding factors that might affect sentiment change. Even with both user fixed effect and day-fixed effect included, the effect of social learning remains highly significant.

Column 4 adds additional controls to address alternative explanations for the effect of social learning. The first story concerns the common news shocks that arrive between post[0] and post[1], echoing the findings in Feng and Seasholes (2004). To account for this, we include the contemporaneous average

[6]This is formally known as the Simpson's Paradox, a statistical phenomenon where a trend appears in different groups of data, but reverses or disappears when the groups are combined. The paradox arises because the aggregated data may obscure or mask the underlying patterns in the subgroups. It highlights the importance of carefully considering and accounting for potential confounding variables when analyzing and interpreting data, as failing to do so can lead to erroneous conclusions.

[7]For ease of presentation, we scale up each coefficient by a factor of 100.

[8]For example, see Barber and Odean (2001), D'Acunto et al. (2019), Grinblatt et al. (2011).

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news sentiment from Ravenpack News Database between the consecutive posts. We also include lagged Ravenpack news shocks up to 48 hours[9] before post[0] as additional controls to address the concern that investors might respond to news with a delay. Furthermore, we compute the contemporaneous Bitcoin return and volatility from hourly Bitcoin market data between t_0 and t_1, and include them as additional controls for contemporaneous information sources. We find that the regression coefficient of social sentiment remains unchanged. Finally, we construct the forum sentiment variable that measures the average sentiment of posts published between t_0 and t_1 in other conversations than the ones of post[0] or post[1]. Even controlling for all above confounding factors, the coefficient of social sentiment remains significantly positive.

In columns 5 and 6, we test whether the effect of social learning is significant even within user-week or user-day. For example in column 6, by adding user-day fixed effects, we solely rely on variation between different post pairs by the same user on the same day. Even in this stringent specification, the effect of social learning is still significantly positive, suggesting that our results are unlikely driven by time-varying user characteristics. Note that column 6 is based on a much smaller sample than the other columns because we focus on user-days with at least two pairs of posts. In the rest of the paper, most of our analysis is based on the sample of column 4 or column 5.

3.4 Propagation on the Network

We further investigate the propagation of sentiment through the social network. We consider two users as neighbors if they have interacted by participating in the same conversations. Our hypothesis posits that the sentiment of a neighbor's neighbor may affect a user's sentiment revision due to diffusion of sentiment. To test this hypothesis, we trace the conversations of these neighbors within a 24-hour window prior to the current conversation, excluding conversations in which the user herself has previously participated. We refer to these as neighboring conversations. We then calculate the average sentiment of the neighbors' neighbors in these indirectly connected conversations, which we term as indirect social sentiment. Table 3 shows that indirect social sentiment has a positive and significant effect on a user's sentiment, consistent with the hypothesis that a user' sentiment can propagate through a social network and affect other users without direct interactions.

[9] In untabulated results, we also control for news arrivals in the past 7 days and our findings remain consistent.

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3.5 Placebo Test

To further establish the importance of social interactions for sentiment revision, we provide a placebo test. Specifically, we examine whether sentiment changes are affected by the average sentiment in a randomly selected conversation not participated by the user occurring between the timestamps of post[0] and post[1].

Table 4 presents the results. Although columns 1 and 2 find significant effects of other random conversations on the same day, the effect goes away once we control for day fixed effects, add time-varying controls or include user-day fixed effects as in columns 3 to 5.

3.6 Heterogeneity by User Features

We investigate heterogeneity across different subgroups in our user sample. First, classical asset pricing theories assume there are two types of users: naïve users who are often driven by psychological factors and sophisticated users who serve as the counteracting force to market sentiment (De Long et al. (1990), Lee et al. (1991), Barber and Odean (2013)). We test whether naive users are more susceptible to sentiment conveyed through conversations by relying on rankings assigned to users by the forum. On Bitcointalk, users are classified into legendary and non-legendary categories based on their experience and contribution to the Bitcointalk community. We define users in the non-legendary category as naive users, while treating legendary users as sophisticated users. Column 1 of Table 5 studies the heterogeneous effect of social learning on naïve and sophisticated users. The interaction term between a naïve dummy and social sentiment has a positive and statistically significant coefficient, indicating that naive users are much more responsive to sentiment in conversations than sophisticated users are. Compared to a sophisticated user, a naive user updates her sentiment on average by 3.585% more after participating in conversations with a social sentiment of 1.

Second, we compare users who are more central in the social network with other users. We treat two users as connected if they have participated in at least one conversation together. The network structure is updated daily. Following an established algorithm in the network literature (Hagberg et al. (2008)), we calculate the centrality score for each user and define those more connected as *central users*. These central users engage in a broader range of conversations and are likely to be more experienced. Column 3 of Table 5 central users are significantly less responsive to social sentiment. A one-unit increase in the

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centrality score reduces the magnitude of sentiment contagion by 0.047%. This reduction may be attributed to their experience in social interactive activities. However, it is important to note that experience in social interactions moderates but does not always fully eliminate social learning. For instance, for a user at the 75th percentile in terms of centrality score at 16.3, the effect of social learning is still 3.333 (calculated as 4.100 - 0.047 * 16.3).

Third, we study whether users who are less informed respond more to social sentiment. We define less informed users on a rolling basis as those whose sentiment exhibits a low correlation with future Bitcoin returns in 2 days (correlation < -0.2). We conjecture that less informed users may be more influenced by social sentiment. Column 5 of Table 5 presents evidence consistent with this hypothesis. The interaction term indicates that compared with more informed users, less informed users update sentiment by 5.137% more after participating in a conversation with a social sentiment of 1. In columns 2, 4, and 6, we replace user and day fixed effects with the user-week interactive effect and our results remain unaffected.

3.7 Time-Varying Social Learning

The intensity of social learning may vary across time. To understand the time series variation in social learning, we construct three time varying variables that could potentially affect social learning. These tests are designed to shed light on the rationality of social learning. First, we investigate whether social learning is driven by information flow. We identify a day as informative if the total number of news arrivals in the RavenPack News database exceeds its sample median. We expect to see a stronger effect of social learning on informative days if social interactions are largely exchange of information. Column 1 and 2 of Table 6 present the result. The statistically insignificant coefficient on the interaction term suggests that social learning is not stronger when there is more new information. In our untabulated analysis, we alternatively construct informative days solely relying on news labeled as "novel" in the RavenPack database to avoid double counting and our findings remain the same. Therefore, social learning does not seem to reflect information flow.

Second, we investigate whether social learning is influenced by the level of uncertainty in the Bitcoin market. We construct two variables to reflect uncertainty. The first is the standard deviation of news sentiment

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on Bitcoin from the RavenPack News database[10] and second one is the standard deviation of hourly Bitcoin returns within a 24-hour window, calculated on a rolling basis. Columns 3 to 6 of Table 6 show that social learning is significantly stronger in more uncertain times.[11]

4 Informativeness of Social Sentiment

4.1 Social Sentiment and Future Returns

To further understand the rationality of social learning, we estimate the relationship between social sentiment and future returns by running the following specification[12]:

$$\begin{aligned} \text{Cumulative Return}_{i,j,t_1+1\to t_1+k} &= \beta_1 \text{Social Sentiment}_{i,j,t_0\to t_1} \\ &+ \text{Prior Sentiment}_{i,t_0} + \gamma \text{Control}_{t_0\to t_1} + \alpha_i + \gamma_{d(t_0)} + u_{i,t_1}, \end{aligned} \quad (2)$$

where the dependent variable, denoted as $\text{Cumulative Return}_{i,j,t_1+1\to t_1+k}$ represents the ex-post cumulative return in a forward-looking window from hour 1 to hour k after user i has participated in a conversation j that ends at timestamp $t-1$. We study five different time windows, 6, 24, 48, 72 and seven days after the conversation. The main explanatory variable, $\text{Social Sentiment}_{i,j,t_0\to t_1}$, represents the social sentiment encountered by user i in conversation j between timestamps t_0 and t_1. To address potential confounding factors, we further include user (α_i) and day ($\gamma_{d(t_0)}$) fixed effects, a set of time-varying controls. Standard errors are clustered by user and day, as recommended by Hodrick (1992).

Table 7 shows that social sentiment is largely misinformed. Social sentiment negatively predicts future

[10] It is represented as a binary variable that takes on a value of 1 if the standard deviation is higher than the sample median, and 0 otherwise.

[11] Tversky and Kahneman (1974) establish that availability heuristics influence judgments made under uncertainty.

[12] This specification follows a well-established approach in the retail trading literature (Barber and Odean (2000)). The retail trading literature assesses the quality of decision-making by retail traders based on the timing of buys and sells relative to subsequent returns. Similarly, we evaluate the quality of information linked to social sentiment by analyzing its timing in relation to subsequent Bitcoin returns. In this context, a positive coefficient for β_1 would indicate that social sentiment is informative and enhances the quality of users' signals. Conversely, a negative coefficient indicates low informativeness, suggesting that social sentiment undermines the quality of users' signals.

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returns in most horizons. If users follow a trading strategy of buying (selling) bitcoin immediately after observing positive (negative) social sentiment, on average they would incur a total loss of -0.094%. In comparison, Table 1 shows that a simple buy-and-hold strategy yields a daily return of 0.326% (calculated as 1.191/365).

4.2 Social Learning Through Informative Posts

While a higher social sentiment on average predicts negative returns, there may well be informative users whose posts positively forecast returns. To explore this hypothesis, we use three user features studies in 5: level of being informed, centrality, and legendary status. We decompose social sentiment into two parts for each selected feature: social sentiment by featured users and non-featured users. We then investigate whether each component predicts future returns.

Panel A of Table 8 presents our findings. Column 1 shows that social sentiment from informed users more positively predicts returns, even though the coefficient in column 1 is not statistically significant. In contrast, social sentiment from non-informed users negatively predicts future returns. Columns 3 and 4 demonstrate that the social sentiment of both central and non-central users negatively predicts returns. Somewhat surprisingly, social sentiment from central users more strongly predicts negative returns than social sentiment from non-central users. In columns 5 and 6, we compare the informativeness of social sentiment from legendary and naive users. Social sentiment from naive users is slightly less informative than that from legendary users.

In Panel B of Table 8, we further investigate the response of users to different components of social sentiment studied in Panel A. Columns 1 and 2 reveal that users exhibit a stronger reaction to social sentiment from non-informed users. This finding implies that such a social learning pattern is unlikely to be rational, considering the negative predictive power of social sentiment from non-informed users. In columns 3 and 4, we can observe that users strongly respond to the social sentiment of both central and non-central users. However, the social sentiment from both types of users negatively predicts future returns. In columns 5 and 6, we find that users are more sensitive to social sentiment from naive users, responding twice as much as to legendary users. This is irrational, given that columns 5 and 6 in Panel A show that sentiment of naive users is more likely to predict negative returns than sentiment of legendary users. In untabulated tables, we

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include the user-week interactive fixed effect and find our results remain similar.

In summary, the findings from Panel A and B of Table 8 lend additional support to the behavioral narrative of social learning: different components of social sentiment are generally uninformative, but users actively learn from them and revise their sentiment accordingly.

5 Confirmation Bias

Why would users respond irrationally to social sentiment? The psychology literature has extensively documented humans' cognitive limitations, which affect our perception, attention, memory, decision-making, and problem-solving abilities. One notable heuristic is confirmation bias (Nickerson (1998)), the tendency to search for, interpret, favor, and recall information in a way that confirms or supports one's prior beliefs or values. In this section, We study confirmation bias in the process of sentiment contagion. We provide two pieces of evidence consistent with confirmation bias. First, users selectively participate in confirmatory conversations, a phenomenon also known as *echo chambers*. Second, they also selectively interpret the signals they encounter in conversations.

5.1 Echo Chambers: Selective Participation of Conversations

Confirmation bias may cause users to selectively acquire signals. We examine users' decisions to participate in conversations that are confirmatory with their priors by estimating the following probit model:[13]

$$\begin{aligned} & Pr[\text{Participate Positive}_{i,t+k} = 1]_t \\ = \; & \Phi[\beta_0 + \beta_1 \text{Prior Sentiment}_{i,t} + \gamma_m \text{Control}_{i,t,m} + u_{t+k}], \end{aligned} \tag{3}$$

where Φ is the $c.d.f.$ of the standard normal distribution. The dependent variable Participate Positive is an indicator, taking a value of 1 if the subsequent conversation has a positive sentiment and 0 otherwise. We employ two measures for whether the subsequent conversation has a positive sentiment. The first measure is based on whether the first post of subsequent conversation has a positive sentiment. We focus on the first post because it garners the most attention when users browse through threads. The second measure is social sentiment, setting the dependent variable to 1 if the social sentiment is positive and 0 otherwise.

[13]The consecutive pairs of posts enable us to observe both users' prior beliefs and their subsequent participation decisions.

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The explanatory variable, Prior Sentiment$_{i,t}$, is an indicator for whether user i's sentiment is positive in her post published at the timestamp of t. We also include a battery of control variables Control$_{i,t,m}$ to account for market and forum events that occur between the two consecutive timestamps t and $t+k$. The control variables are identical to those in Table 2.

Echo chambers predict that users with positive priors would deliberately choose to participate in conversations signaling positive sentiment. We would therefore anticipate a positive coefficient for the Prior Sentiment$_{i,t}$ from estimating equation (7). The marginal effects reported in Table 9 corroborate the existence of echo chambers. For example, in column 1, the interpretation of the reported marginal effects is as follows: users with positive priors, on average, are 6.734% more likely to subsequently participate in a thread that starts with a post with positive sentiment, compared to users with non-positive priors. The unconditional probability of participating in a conversation that starts with a positive sentiment is 37.310%. Thus having a positive prior increases the probability of participating in a thread with a positive first post by 18.049% $(6.734/37.310)$. The marginal effects of positive priors are robust to additional controls added in column 2. In columns 3 and 4, we estimate the regression with an alternative dependent variable that equals 1 when the social sentiment is positive and 0 otherwise. The marginal effects of positive priors remain significant at the 1% level.

5.2 Selective Interpretation of Information

Confirmation bias can lead users to interpret information in a way that confirms their priors. To examine this phenomenon, we investigate how users selectively interpret signals they encounter in conversations. Specifically, we create two new variables: social sentiment (-) and social sentiment(+). Social sentiment (-) is defined as the minimum value of social sentiment and zero. Social sentiment (+) is defined as the maximum value of social sentiment and zero. These two variables capture the linear relationship between sentiment contagion and social sentiment in the negative and positive regions, respectively. We then conduct two subsample analyses to study selective interpretation: one for users with positive priors and one for those with non-positive priors. Confirmation bias predicts that users with positive priors would respond more strongly to social sentiment (+), while users with negative priors would respond more aggressively to social sentiment (-).

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Table 10 presents our findings. In column 1, we observe that users with positive priors adjust their sentiment towards the social sentiment, which supports the main message conveyed in Table 2. Column 2 shows that users with positive priors respond to both positive and negative social sentiment, but they are more responsive to positive sentiment. Specifically, a one-unit increase in the positive sentiment region is associated with a 5.006% probability of a sentiment upward change, which is roughly 82.567% higher than the probability of responding to increases in the negative sentiment region (although this difference is not statistically significant). In column 3, we find that users with negative priors also adjust their sentiment towards the social sentiment. Finally, in column 4, we observe that users with negative priors respond to both positive and negative social sentiment, but they are more responsive to negative sentiment. Specifically, a one-unit increase in the negative sentiment region is associated with a 4.250% probability of a sentiment upward change, which is roughly 51.623% higher than the probability of responding to increases in the positive sentiment region. In untabulated tables, we include the user-week interactive fixed effect and find our results remain similar. Together, results in Table 10 indicate a selective interpretation of information: investors respond more aggressively to the social sentiment that is consistent with their prior sentiment. In untabulated tables, we include the user-week interactive fixed effect and find that our results remain similar. Taken together, the results in Table 10 indicate a selective interpretation of information: investors respond more aggressively to the social sentiment that is consistent with their prior sentiment.

6 Social Interactions, Individual Trading, and Market Outcomes

How does sentiment contagion from social interaction impact the Bitcoin market? In this section, we explore the impact of social interactions on individual trading decisions and the overall market.

6.1 Social Learning and Individual Trading

We begin by discussing how we measure individuals' trading decisions. Out of the 37,262 users in our sample, 1,284 users voluntarily published their Bitcoin wallet address following the recommendation of the Bitcointalk website (explained in Section 2.3). Since Bitcoin transactions are public on chain, we have access to the complete transaction history of these 1,284 users. We identify sentiment changes when users publish two consecutive pairs of posts and aggregate all trades that occur between these posts. To evaluate

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the predictive power of sentiment changes on trading, we also aggregate trades within a subsequent 7-day window after the publication of post[1]. Similar to regression 1, the unit of observation is the time window between two consecutive pairs of posts by the same individual. To establish the micro-level link between sentiment change and individual trading, we regress trading decisions for individual i on the sentiment change between post[0] and post[1] in conversation j, denoted as $SentiChange_{i,j,t_0 \to t_1}$ (t_0 and t_1 are the timestamps of post[0] and post[1]):

$$\text{Trade}_{i,j,t_0 \to t_1} = \beta \text{Senti Change}_{i,j,t_0 \to t_1} + \gamma \text{Control}_{i,j,t_0 \to t_1} + \text{Fixed Effects} + u_{t+k}. \tag{4}$$

The vector $Control_{i,j,t_0 \to t_1}$ consists of control variables used in regression (1). Additionally, we include the time gap between t_0 and t_1 as an additional control, as trading is more likely to occur over longer time windows. In all specifications, we include both user and date fixed effects to subsume unobserved user- and date-level variations. We cluster standard errors at user and date levels.

We report the results of regression equation (4) in Table 11. Columns 1 and 2 investigate if sentiment change unconditionally explains individuals' trading decisions. Since the dependent variable is non-directional, we use the absolute value of sentiment change as the explanatory variable. In column 1, the dependent variable is an indicator of whether the user has traded at least once between the time gap of post[0] and post[1]. The ability of sentiment change to unconditionally predict trading between t_0 and t_1 is not statistically significant. In column 2, the dependent variable measures whether the user has traded at least once within a 7-day window after post[1]. This represents a predictive regression since the time horizon of the dependent variable does not overlap with sentiment change. We find a similar pattern where sentiment change does not unconditionally predict individuals' future trading decisions.

The low sensitivity in columns 1 and 2 may indicate two distinct mechanisms. Firstly, it could be a consequence of infrequent trading, resulting in an overall low responsiveness to changes in sentiment. Alternatively, it could signify a limited sensitivity of transactions to sentiment changes, even with active trading by users. We investigate these explanations in columns 3 to 5.

We condition on time gaps during which individuals trade actively, and study if the direction of trade could be explained by sentiment changes. Column 3 presents the results of a regression similar to column 1, with the dependent variable now indicating whether users have increased their holdings by at least 0.1 Bitcoins during the window $[t_0, t_1]$. It is important to note that trading frequency is very low, accounting for

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only 1.066% of our sample, which explains the small number of observations in columns 3 and 4. However, when trading does occur, we find that sentiment changes are predictive of the trading direction. Specifically, a one-unit increase in sentiment change increases the likelihood of actively increasing Bitcoin holdings by 7.011 percentage points within a given window. This economic effect is more than six times larger than the unconditional trading frequencies (7.011/1.066 = 6.577). In column 4, we further impose a restriction that the time window must be longer than 2 hours to allow users sufficient time to place their trade orders. We discover that the coefficient of sentiment change increases by 31.850% (calculated as 9.244/7.011 - 1), indicating that a longer window indeed strengthens the translation of sentiment changes into trades.

In column 5, we analyze the influence of sentiment changes on individuals' future trading decisions within a 7-day window following post[1], given that trading takes place. The percentage of windows leading to future trades within this 7-day period is 17.686%, significantly higher than the trading frequency within the $[t_0, t_1]$ window. When trades occur within the 7-day window, a one-unit increase in sentiment changes raises the probability of actively buying Bitcoins by 0.576 percentage points. This coefficient is statistically significant. However, the economic impact is smaller compared to predicting trades within the $[t_0, t_1]$ window, as future trading decisions are influenced by other factors unrelated to sentiment changes. The findings in columns 3 to 5 demonstrate a strong connection between sentiment changes and trading among users. The lack of significance in the unconditional explanatory pattern is primarily attributed to low trading frequencies.

In summary, this observation establishes a micro-foundation for understanding the influence of social interactions on market outcomes. By combining the findings presented here with those in Table 7, we further conclude that social sentiment has a negative impact on users' performance, leading to underperformance.

A recent study by Giglio et al. (2021) also examines the impact of belief changes on trading decisions. Their study finds that belief changes cannot predict trading occurrence (extensive margin) due to infrequent trading. However, belief changes can explain the direction of trading after a transaction is made (intensive margin). Our results align with their findings. While their study examines belief changes every two months, we are able to observe changes in sentiment at high frequency, thus providing additional supportive evidence. They focus on wealthy users who make cautious decisions, while we study highly sentimental users. Moreover, we extend their conclusions by analyzing not only equity but also cryptocurrency asset

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classes. What further distinguishes our findings from theirs is the fact that individuals' trading decisions can be inspired by social interactive activities. We uncover this novel link by leveraging the unique features of the Bitcoin market – the fact that the data on social interactive activities and trading decisions are available simultaneously, and investors have sufficient motivation to engage in social interactions to gather information to inform their trading decisions.

6.2 Social Learning and Market Outcomes

The link between sentiment changes and trading decisions discussed in section 6.1 provides a foundation for studying how social interactions affect market outcomes. We start by constructing a daily Sentiment Contagion Intensity (SCI) index to capture the intensity of sentiment contagion in social interactions. Unlike previous studies, SCI captures user sentiment dynamically, reflecting the intensity of contagion rather than the average static level. SCI has another unique feature: it is built on social interactions. This enables us to examine how social interactions affect market outcomes through sentiment contagion.

6.2.1 Construction of SCI

To establish SCI, we follow two steps. First, we focus on users whose sentiment changes after participating in a conversation. To ensure that these changes are induced by social interactions, we restrict the direction of sentiment change to match the social sentiment in the conversation. We count the total number of affected users within a given day, which reflects the intensity of sentiment contagion.

One caveat is that an increase in affected users may be driven by the growing number of participants on the Bitcointalk website. As Bitcointalk gains reputation, the total number of active users on the forum may also rise. To address this, in our second step, we remove the time trend and seasonality by regressing the series for affected users on a weekday and year-month-pair indicators. The resulting residual is our SCI index. It has a mean of zero and a standard deviation of 0.968.

6.2.2 SCI and Future Volume

In light of the link between changes in sentiment and individuals' trading decisions, we put forth a hypothesis stating that the SCI index is a positive predictor of future trading volume in the Bitcoin market.

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Specifically, on days when the SCI is high, we expect to see a greater number of users engaging in trades in the coming days, as they adjust their sentiment based on social interactions.

To examine our hypothesis, we conduct a regression analysis using the following equation:

$$\text{Ab Volume}_{t+N} = \beta_0 + \beta_1 SCI_{i,t} + \Sigma_m \gamma_m \text{Controls}_{t,m} + u_{t+k} \tag{5}$$

where Ab Volume_{t+N} denotes the abnormal trading volume of Bitcoin on the next N days. Specifically, we normalize the average trading volume over the next 1 day and 7 days ($N = 1, 7$) by subtracting the mean level of trading volume in the past 14 days. The key predictor is $SCI_{i,t}$. As for the control variables, $\text{Controls}_{t,m}$ encompasses events occurring within the last 14 ($m = 14$) days. To proxy for news arrivals, we calculate the average sentiment levels in Ravenpack news during the previous 14 days. Additionally, we control for Bitcoin Volatility and the Number of Transactions, which consider the cumulative sum within the past 14 days. To account for forum sentiment, we calculate the average sentiment of posts on the Bitcointalk forum published within the previous 14 days.

The findings presented in columns 1 and 2 of Table 12 strongly support our hypothesis. Overall, we observe a significant association between an increase in SCI and abnormal trading volume in the subsequent days. For instance, in column 1, a one standard deviation increase in SCI leads, on average, to an additional 3.884 million units of abnormal trading volume the following day (calculated as 401.246/1000.968). Importantly, even after controlling for news shocks, market fluctuations, and forum sentiment over the past 14 days, this relationship between SCI and abnormal trading volume remains unaffected. In column 2, we report the predictive power of SCI for the average abnormal trading volume over the next 7 days. We find that a one standard deviation increase in SCI results in an additional 2.945 million units of abnormal trading volume per day over the next 7 days (calculated as 401.246/1000.968).

6.2.3 SCI and Future Volatility

Extensive research (Black (1986)) has long been devoted to examining the connection between user sentiment and asset price volatility. If users base their trading decisions on sentiment, then changes in sentiment lead to more noise trading and generate excessive volatility. Notably, recent studies (Da et al. (2014), Antweiler and Frank (2004)) have provided empirical evidence supporting this relationship, utilizing data from the U.S. stock market. However, none of these studies specifically emphasize the role of social

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interactions in this context. The SCI indicator naturally helps fill the gap. Specifically, we hypothesize that SCI index would positively predict future market volatility: on days when the SCI is high, a greater number of users would be affected by social interactions and engage in trades in the coming days, pushing up future volatility.

To examine this conjecture, we conduct the following regression analysis:

$$\ln(rv_{t+N}) = \beta_0 + \beta_1 SCI_t + \Sigma_m \gamma_m \text{Control}_{t,m} + u_{t+N} \tag{6}$$

rv_{t+N} represents the future realized volatility. To calculate it, we compute the standard deviation of hourly returns for each day over the next N days. We then take the average of these values and transform it into an annualized unit. We take $N = 1, 7$. The key explanatory variable of interest and the control variables are identical to those in equation 5.

The results of our study are presented in columns 3 and 4 of Table 12. We observe a positive and significant relationship between the future volatility of Bitcoin returns in the next few days and the SCI indicator. In column 3, a one standard deviation increase in SCI corresponds to a 0.519% (calculated as 0.536/100*0.968) increase in realized volatility over the next day. This increase amounts to 14.84% (calculated as 0.519%/3.496%) of the standard deviation of the realized volatility. In column 4, we examine how SCI predicts the average abnormal trading volume over the next 7 days and find a similar pattern. Overall, results in columns 3 and 4 are consistent with our conjecture: as social interactions trigger more sentiment changes, there is an increase in noise trading by users, which subsequently raises the volatility of returns in the market.

6.3 Socially Constructed Optimism: Impact on Market Crashes and Volatility

In David Hirshleifer's presidential address (Hirshleifer (2020), he pointed out an important feature corresponding the impact of social activities on the market: error-prone. As sentiment percolates through populations, more users may be influenced by social interactions and become sentimental. This accumulated optimism can destabilize the market by driving asset prices too high initially and then causing a downward correction as sentiment reverts back to normal. Therefore, in this section, we empirically investigate whether such accumulated optimism predicts a higher probability of market instability. We measure optimism in forum discussions and link it to future market dynamics, specifically market volatility and the occurrence of

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market crashes.

We construct two measurements for optimism at the hourly level. The first measure, which we term *optimism in summation*, quantifies optimism by summing the sentiment scores across posts published within each hour. A higher value for optimism in summation indicates a greater prevalence of optimistic views being expressed by users in the forum. The second metric, known as the *optimism in fraction*, quantifies the proportion of hours within the previous 7*24 hours that exhibit positive aggregated sentiment. The aggregated sentiment is derived from the summation of sentiment scores across posts published within each hour. A higher value of this metric indicates a stronger inclination among users to express positive sentiments during the preceding week. To facilitate interpretation, in our subsequent analysis, two measures are both expressed in standard deviation units by standardizing with the mean and standard deviation.

By estimating the following probit regressions, we first examine whether socially constructed optimism predicts increased crash risk in the Bitcion market:

$$Pr[\text{Crash}_{t+k} = 1]_t = \Phi[\beta_0 + \beta_1 \text{Optimism}_t + \gamma_m \text{Controls}_{t,m} + u_{t+k}], \tag{7}$$

where Φ is the $c.d.f.$ of the standard normal distribution. The dependent variable $\text{Crash}_{t+k} = 1$ is a future crash indicator in the Bitcoin market, which takes on a value of 1 if there is any return crash in the next K hours (K =6, 24, 48, and 72) and 0 otherwise. We classify the event of return crash in a given hour if the future Bitcoin return within the subsequent 24 hours is less than -15%. The socially constructed measures of optimism serve as the key explanatory variable. Table 13 presents the marginal effects of crashes in the Bitcoin market, conditioned on a one standard deviation increase in socially constructed optimism. The columns in Table 13 represent the increased probability of a crash event ahead by 6, 24, and 72 hours. Furthermore, we consider other factors such as RavenPack News sentiment and market dynamics, including cumulative return and return volatility from the previous week.

Table 13 demonstrates that socially constructed optimism is positively correlated with an increased likelihood of crashes. Taking optimism as a whole over a 24-hour time horizon as an example, the interpretation of the reported marginal effects is as follows: a one standard deviation increase in optimism is associated with a statistically significant 0.419% increase in the probability of a subsequent crash in the Bitcoin market at a 5% level of significance. As a point of reference, the unconditional probability of a crash event within the next 24 hours is 2.741%. Thus, a one standard deviation increase in socially constructed optimism raises

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the probability of a crash event by approximately 15.286%. Alternatively, if we focus on optimism as a fraction within a 24-hour time horizon, a one standard deviation increase in socially constructed optimism results in an approximate 43.123% (1.182/2.741) increase in the probability of a crash event.

We also establish a link between the socially constructed measures of optimism and future market volatility. Market volatility is calculated based on hourly returns over a specified time horizon (K = 6, 24, and 72 hours). Our findings in Table 14 reveal a strong and positive predictive pattern. Specifically, a one standard deviation increase in the measure of optimism results in an 0.419 percentage point increase in future volatility within the next 24 hours (column 3), when considering the summation measure of optimism. This predictive pattern remains robust even after controlling for factors such as Ravenpack News, cumulative Bitcoin returns, and Bitcoin volatility over the past 7 weeks. A similar pattern is observed when analyzing the measure of optimism in fraction, although the magnitude of the coefficient drops to 1.182 percentage point (column 4). Furthermore, our analysis demonstrates that this predictive pattern persists when extending the time horizon to future volatility in the next 24 or 72 hours.

7 Social Learning and Bubbles

Bubbles have long been recognized as episodes featuring elevated investor sentiment and irrational exuberance, along with rapid rise in both asset prices and trading volume. Researchers are fascinated by these features and the underlying mechanisms behind them. The Bitcoin market provides a valuable opportunity to investigate bubbles as it has been often associated with bubbles since its inception. In the past decades, Bitcoin prices have experienced several drastic changes not easily explainable by fundamentals. Moreover, Bitcoin was born in an age with various developed social platforms, and the process of its popularity is also closely related to the social network. Our data allows us to provide evidence on the time-varying feature of social interactions and sentiment contagion during bubble episodes.

7.1 Identify Bitcoin Bubbles

We identify bubbles by focusing on days when the Bitcoin price rapidly rises to local peaks. Specifically, we select days with a cumulative Bitcoin return higher than 200% over the past quarter. For each set of consecutively identified days, we group them together in preparation for a bubble episode. Within each

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episode, we consider the day with the highest Bitcoin price as the peak day and the day two quarters prior to the peak day as the start date. The bubble episode encompasses the period between the start date and the peak date. [14]

Our choice of 200% return is intended to conform to Fama and others' notion that a bubble, if it does exist, is associated with a substantial price run-up. A return threshold of 200% effectively captures the majority of episodes that anecdotal evidence suggests were Bitcoin bubbles, including the episode towards the end of 2017. It is worth noting that we define bubbles ex-post by relying on future information. This approach does not undermine our analysis since we do not aim to predict bubbles.

We identify four bubble episodes within our sample horizon. The first one starts on June 4, 2013, and extends to December 4, 2013. The second one begins on June 16, 2017, and ends on December 16, 2017. The third one spans from December 26, 2018, to June 26, 2019. The fourth one starts on September 13, 2020, and continues until March 13, 2021.

In Table 15, we present key features of the identified bubble episodes and compare them to the non-bubble episodes in the Bitcoin market. The differences in features between the bubble and non-bubble episodes are statistically significant, as supported by the joint F-statistic of 15.49 obtained from seemingly unrelated regression analysis. In Panel A, we provide summary statistics for market variables. We observe that, in the identified bubble episodes, the daily Bitcoin return (annualized) is more than 8 times higher (calculated as 4.050/0.491) compared to the non-bubble episodes, and the return volatility within a day also increases by 8.108% (calculated as 0.040/0.037 - 1). The total dollar volume nearly doubles during the bubble episodes. When considering news reports, the RavenPack news sentiment rises from 0.040 to 0.168, which is more than four times higher than the sentiment level in the non-bubble episodes. Moreover, abnormal Google Search volume surges from 0.017 to 0.085. To summarize, the identified episodes exhibit rapid increases in returns, volume, and market fluctuations. Furthermore, there is a noticeable prevalence of optimism in media coverage, along with a surge in investor attention. All these characteristics align with the commonly described attributes of a bubble (Shiller (2001)).

[14] A window of two quarters well captures most Bitcoin bubble episodes. Our results are robust to alternative windows.

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7.2 Social Interactions in Bubbles

7.2.1 Overall Description

Our Bitcointalk data provides valuable insights into the notable characteristics of social interactions during bubbles. Sharp changes in market conditions would plausibly impact conversation patterns concerning Bitcoin in various ways, including shifts in content and attitudes. In Panel B of Table 15, we summarize the crucial aspects of investor sentiment and social interactive activities during bubble episodes.

When it comes to investor sentiment, it is unsurprising to observe that users tend to be more optimistic during bubble episodes. The user sentiment increases by 12.132% (calculated as 0.305/0.272 - 1), indicating a pervasive sense of optimism in social sentiment. At the same time, user sentiment becomes less dispersed in bubbles—the standard deviation of sentiment drops by 2.232%. Such a drop in disagreement is remarkable, especially given the 14.831% (calculated as 133.504/116.261 - 1) increase in the total number of users participating in the conversations during bubbles.

In terms of social interactions, there is a significant increase in the total number of posts per day by 7.428% (calculated as 188.764/175.713 - 1). This indicates a heightened intensity in social interactions during bubbles. It is worth noting that the number of posts with positive sentiment increases disproportionately: the fraction of posts with positive sentiment increases by 5.233%. This asymmetric pattern is potentially connected to the self-enhancing transmission bias (Han et al. (2022)).[15] We provide supportive evidence in Table A6 and A7. Moreover, we observe a significant decline of 16.156% in the fraction of sophisticated users during bubbles (calculated as 1 - 0.301/0.359). This suggests that a disproportionately large number of novice investors participate in conversations during bubble episodes. These identified features of social interactions in Panel B generally align with the descriptions of bubble episodes in Shiller (2001).

7.2.2 Elevated Contagion of Optimism

The salient features of social interactions may further reshape social learning in bubbles. As evidenced in Panel B of Table 15, the change in the fraction of positive and negative sentiment is asymmetric. To emphasize the difference in sentiment contagion between optimism and pessimism during bubbles, we present

[15] Days in bubbles tend to exhibit high past returns, which in turn increases the likelihood of investors publishing posts with positive sentiment.

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two distinct measures in Panel C of Table 15. Specifically, we identify users who become more optimistic after encountering positive social sentiment as *positively infected users*, following the same procedure outlined in Section 6.2.1 when constructing the Social Contagion Index (SCI). Similarly, we define *negatively infected users*. We use the number of positively and negatively infected users each day to investigate social learning in two directions during bubbles.

Our findings reveal a robust propagation of optimism and a diminished propagation of pessimism during bubbles. Compared to a day in non-bubble episodes, the number of positively infected users significantly increases by 8.927% (calculated as 16.704/15.335 - 1) in bubbles. In contrast, the number of negatively infected users on average experiences a significant decrease of 15.998% (calculated as 0.61/3.813). Collectively, the overall propagation of optimism shows a notable increase of 24.925% within one day during bubble episodes.

The notable features of social interactions in bubbles (Panel B of Table 15) provide valuable insights into the pattern. Firstly, users would frequently run into conversations with an overall positive sentiment in bubbles, as social sentiment in these periods exhibits pervasive optimism. Secondly, during bubbles, there is an increasing participation of naive investors in discussions. As we have discussed in section 3.6, these investors are more susceptible to social sentiment. Additionally, sentiment contagion becomes stronger during periods of high uncertainty. The heightened volatility in bubble episodes further amplifies the spread of optimism.

7.3 Propagation of Optimism and Trading Volume in Bubbles

Why do investors trade so frequently during bubbles? Explaining the heightened trading volume presents an intriguing challenge for researchers.[16] In this paper, we offer a fresh perspective by examining the role of social learning, specifically focusing on the intensified contagion of optimism. This intuition builds upon our previous findings, where we documented that sentiment contagion in social interactions prompts investors to trade, subsequently predicting the trading volume at the aggregate market level. Given that the contagion of optimism intensifies during bubble episodes, it follows that we should expect a surge in trading volume during these periods.

[16] See Barberis et al. (2018), DeFusco et al. (2017), Liao et al. (2021).

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To effectively demonstrate the strong correlation between the propagation of optimism and trading volume in Bitcoin bubbles, a plot serves as the most suitable method. In Figure 4, we observe the temporal trend for both the number of positively infected investors (indicated by the red dashed line) on day t, and the trading volume on day $t+1$ (depicted by the blue bars). The definition of positively infected investors aligns with the approach outlined in section 6.2.1, where we construct the Social Contagion Index (SCI). The trading volume is measured in millions of U.S. dollars. Furthermore, we provide an illustration of the Bitcoin price dynamics (represented by the green solid line) for each identified bubble episode.

We identified four Bitcoin bubble episodes spanning from 2013 to 2021, but for brevity, we will focus on the earliest episode (episode 1) and the most recent episode (episode 4).[17] In both episodes, we observe a synchronous pattern between the number of positively infected users and the trading volume on the following day. In the upper panel, we plot the bubble formation episode from January 2013 to July 2013. When the number of positively infected investors increases, the trading volume on the next day tends to increase sharply. The correlation coefficient between the number of positively infected users and trading volume is statistically significant at 0.669 (p-value = 0.00). This surprisingly high correlation in episode 1 becomes reasonable considering the fact that the Bitcointalk forum was one of the few places where Bitcoin investors could communicate in the early 2010s. In the lower panel, we focus on the most recent bubble-formation episode from September 2020 to May 2021. The correlation coefficient is still statistically positive, but the magnitude drops to 0.335 (p-value =0.00). This may reflect the fact that while Bitcointalk still holds significant influence, it has faced competition from other emerging social network platforms. As Bitcoin investors continue to participate in a more diverse range of social network platforms, the representativeness of Bitcointalk is inevitably decreasing.

In summary, Figure 4 presents compelling evidence illustrating the significant influence of social interactions on high trading volumes during Bitcoin bubbles, particularly through the propagation of optimism. Our results not only offer empirical validation for well-established hypotheses regarding bubbles (Shiller (2001)), but also correspond with contemporary efforts to link social dynamics with bubble phenomena (Hirshleifer (2020), Burnside et al. (2016)).

[17] The patterns in other identified episodes are similar.

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8 Conclusion

Drawing on the calls to "transition from behavioral finance to social finance" (Hirshleifer (2020)) and to gain a better understanding of "the epidemiology of narratives" (Shiller (2017)), this paper presents direct evidence for social learning. Utilizing textual analysis to extract investor sentiment from posts on Bitcointalk, a prominent online investment platform, we establish a robust channel for investors' sentiment on Bitcoin to spread through conversations.

We demonstrate that social learning is inefficient in our context: optimistic sentiment expressed in conversations does not predict higher returns, yet investors respond to them and on average become more optimistic about Bitcoin following social interactions on Bitcointalk. Investors' updating of sentiment shows both the echo chamber effect, where investors selectively acquire information, and confirmation bias, their tendency to selectively interpret information. Naive, less socially connected investors, and less informed investors are more susceptible to the influence of peers' sentiment in social interactions.

Moreover, we find that social interactions exert a substantial impact on individual investors' trading decisions and market outcomes. Although sentiment changes triggered by social interactions do not unconditionally predict probability of transactions, they strongly predict trading directions conditional on trades. Building on this micro-foundation, we also demonstrate that the intensity of daily sentiment contagion significantly predicts future Bitcoin volume and return volatility. Additionally, we develop measures of optimism derived from social interactions to assess the error-prone nature of social activities. Our results indicate that socially constructed optimism measures are significantly and positively related to the probability of future crashes of Bitcoin and its volatility.

Furthermore, we contribute novel insights on the formation of bubbles from the perspective of sentiment contagion. Consistent with Fama's concept of a bubble, we identify episodes displaying prominent bubble characteristics by examining price run-ups and crashes. Simultaneously, we observe widespread optimism and a wave of new investors engaging in social interactions during bubble formation. These two characteristics, combined with the increased uncertainty during bubbles, amplify the contagion of optimism. The contagion of optimism accounts for a significant portion of the variations in trading volume during bubbles. Therefore, social learning provides a fresh perspective for understanding the elevated trading volume during bubbles.

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This paper strongly advocates for the significance of social finance in comprehending investors' decision-making processes and market dynamics. The documented phenomenon of sentiment contagion through conversations not only provides fresh perspectives on investors belief dynamics but also underscores the potential of social interactions in elucidating various financial phenomena. The online investment community serves as an ideal platform for investigating other pertinent questions in the realm of social finance, and we leave these inquiries to future research endeavors.

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Figure 1: Empirical Strategy

Panel A presents an example for our empirical strategy. We show how we measure sentiment changes by calculating the difference between sentiment in post[0] and post[1], and how we measure social sentiment by calculating the average sentiment of others' post in conversations. Each conversation is referred to as a thread with a topic. In the conversation, users can interact with others by posting.

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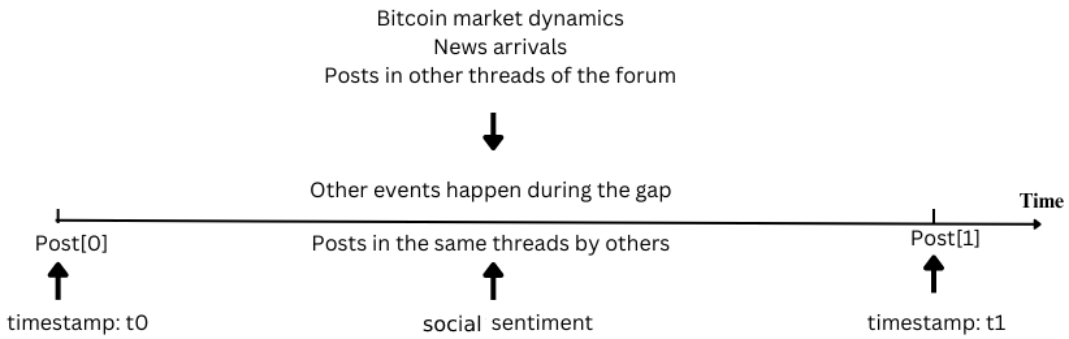


Figure 2: Empirical Strategy

Panel B presents a formal demonstration of our empirical strategy.

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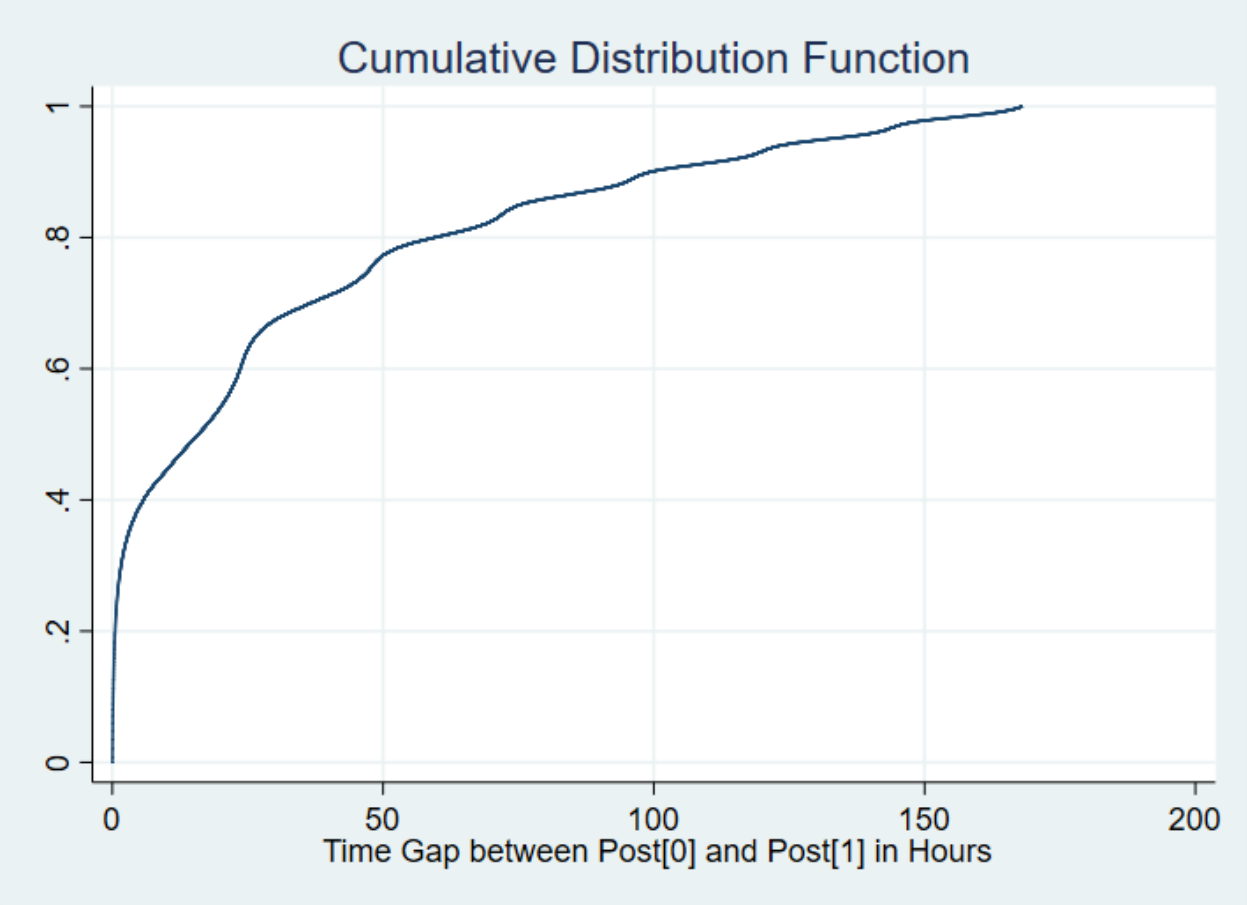


Figure 3: Cumulative Distribution Function of Time Gap

This figure presents the cumulative distribution function of the time gap between two consecutive pairs.

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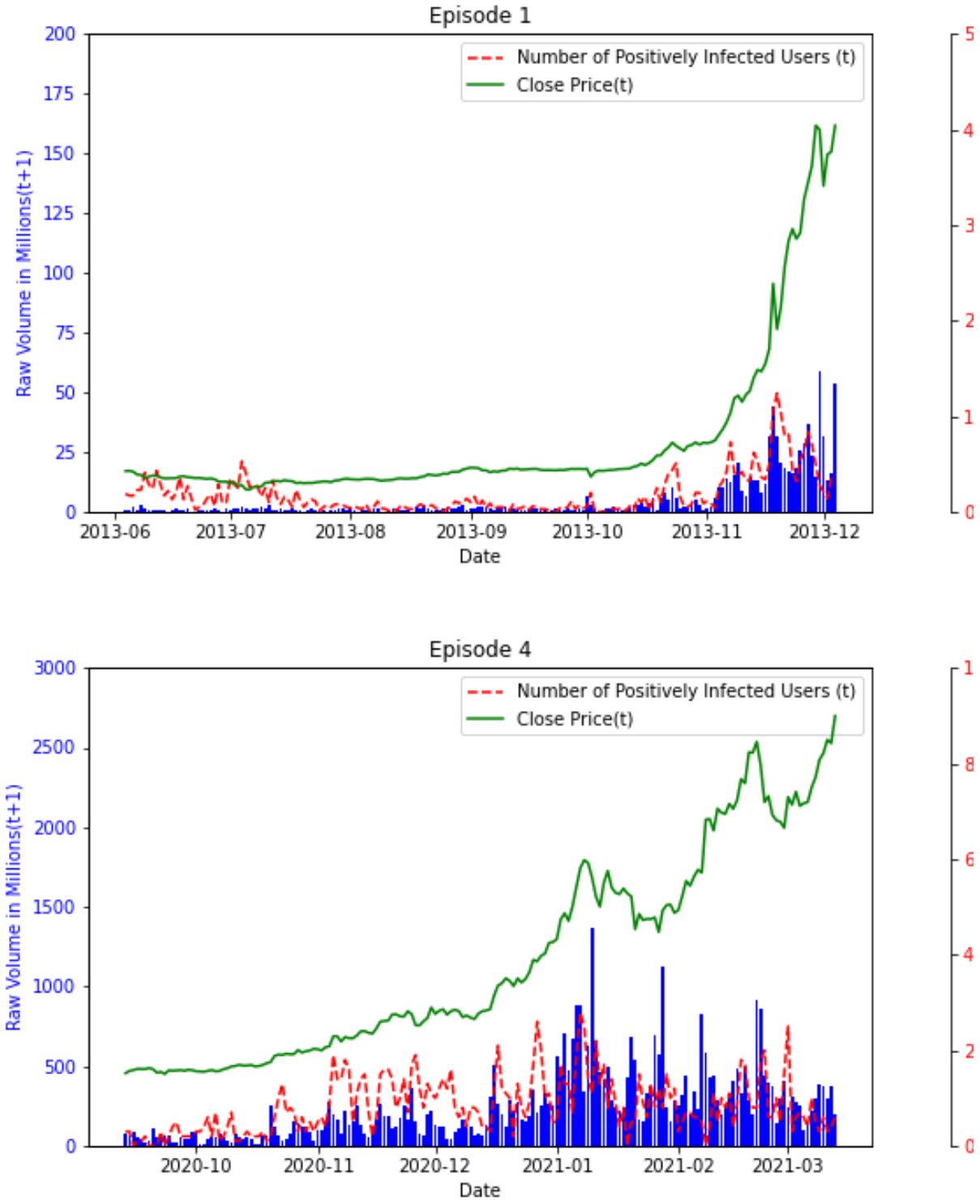


Figure 4: Sentiment Contagion and Trading Volume in Bubbles

This figure plots the number of positively infected users on the day t and the dollar trading volume (in millions) on the next day $t+1$, along with the Bitcoin price index on the day t. We focus on the first (upper panel) and last (lower panel) identified bubbles in our sample.

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Table 1: Summary Statistics

This table tabulates the summary statistics of the key variables. In Panel A, we present features of user sentiment in posts aggregated at the user level. To obtain the summary statistics for the average sentiment, we first calculate the mean sentiment of posts published by each user, then report the corresponding statistics based on the whole user population. In Panel B, we present features of user sentiment in posts aggregated at the daily level. In Panel C, we present features of user sentiment in posts aggregated at the thread level. In Panel D, we present summary statistics of market variables, RavenPack news sentiment and google search volume. Daily Return is the annualized daily return of Bitcoin, Return Volatility (Within Day) is the within-day volatility of Bitcoin based on the hourly Bitcoin return. Number of Transactions is the total number of Bitcoin transactions within one day. Number of Bitcoins Traded is the total number of Bitcoins being traded within one day. Total Dollar Volume (in millions) is the total trading volume of Bitcoin measured in millions of dollars within one day. RavenPack News Sentiment is the sentiment score for the news on Bitcoin in the RavenPack database. We normalize the sentiment score to be between $[-1, 1]$. Our sample spans from Jan 5, 2012 to July 30, 2022.

	count	mean	p50	sd	min	max
Panel A: Post Activities at User level						
Average Sentiment	44,356	0.324	0.333	0.512	-1.000	1.000
Standard Deviation of Sentiment	28,076	0.608	0.641	0.297	0.000	1.414
Number of Posts	44,356	15.015	2.000	72.467	1	3850
Panel B: Post Activities at Daily level						
Average Sentiment	3,736	0.279	0.282	0.118	-0.250	0.833
Standard Deviation of Sentiment	3,735	0.669	0.672	0.054	0.392	0.917
Number of Users	3,736	119.630	85.000	113.392	1	980
Number of Posts	3,736	178.263	128.000	160.777	1	1400
Panel C: Post Activities at Thread level						
Average Sentiment	18,799	0.237	0.250	0.324	-1.000	1.000
Standard Deviation of Sentiment	17,190	0.646	0.659	0.189	0.000	1.414
Number of Users	18,799	24.838	11.000	55.594	1	2589
Number of Posts	18,799	35.427	13.000	551.521	1	73859
Median Gap in Days between Consecutive Posts	17,214	0.573	0.033	16.568	0.000	1129.115
Panel D: Other Data Sources						
Daily Return(Annualized)	3,742	1.191	0.714	16.168	-177.093	146.518
Daily Return Volatility(Annualized)	3,742	0.713	0.555	0.668	0.000	11.450
Number of Transactions	3742	16475.91	11380.50	17269.01	0	181616
Number of Bitcoins Traded	3,742	9361.00	6663.91	9697.85	0	137070.18
Total Dollar Volume (in millions)	3742	56.514	18.862	97.474	0.000	1372.717
RavenPack News Sentiment	3,742	0.041	0.000	0.330	-0.660	0.660

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Table 2: Contagion Effect

This table presents the panel regression analysis of sentiment change on social sentiment. The dependent variable is the sentiment change defined as the revision in sentiment between a user's two consecutive posts post[0] and post[1]. The main explanatory variable is the social sentiment defined as the average sentiment of others' posts within the conversation, published between the timestamps of post[0] and post[1]. We control for simultaneous events between post[0] and post[1], such as Bitcoin return and volatility, activities in other conversations within the forum, and news arrivals documented in the RavenPack database. We also include lagged news arrivals to capture a delay in response. In all columns, we include a prior fixed effect. The t-statistics (in parentheses) are based on standard errors clustered by user and day. We multiply each coefficient by 100.

	Sentiment Change					
	(1)	(2)	(3)	(4)	(5)	(6)
Social Sentiment	8.896***	5.336***	2.844***	3.100***	4.228***	4.018***
	(21.59)	(13.54)	(6.83)	(6.03)	(6.91)	(5.21)
RavenPack News Sentiment				-0.037	1.569*	0.541
between Post[0] and Post[1]				(-0.04)	(1.70)	(0.38)
RavenPack News Sentiment				1.378	1.681*	0.542
24 hours before Post[0]				(1.44)	(1.86)	(0.31)
RavenPack News Sentiment				-1.021	-0.007	-2.109
48 hours before Post[0]				(-1.02)	(-0.01)	(-1.20)
Bitcoin Return				30.727***	32.487***	29.934***
				(5.90)	(5.62)	(4.59)
Bitcoin Volatility				13.020	5.273	-7.379
				(0.81)	(0.30)	(-0.32)
Forum Sentiment				-4.691***	1.623	0.869
				(-4.43)	(1.35)	(0.58)
Prior FE	YES	YES	YES	YES	YES	YES
User FE	NO	YES	YES	YES	NO	NO
Day FE	NO	NO	YES	YES	NO	NO
User X Week FE	NO	NO	NO	NO	YES	NO
User X Day FE	NO	NO	NO	NO	NO	YES
Controls	NO	NO	NO	YES	YES	YES
Adjusted R-Squared	0.338	0.344	0.344	0.345	0.322	0.305
N	218,268	212,647	212,623	175,750	143,092	98,910

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Table 3: Contagion Effect: Propagation on the Network

This table presents the propagation of the contagion effect through the network. Specifically, we track the participation of other users in each conversation (referred to as "neighbors") and examine the sentiment expressed by their neighbors (referred to as "neighbors' neighbors") in previous conversations within the 24-hour window prior to joining the current conversation. Since neighbors' neighbors are indirectly connected to the user through their neighbors, we refer to the average of their sentiment as the "indirect social sentiment.". We are interested in how the indirect social sentiment affects the user's sentiment updating. Therefore, the main explanatory variable is the average sentiment of the neighbors' neighbors, while the control variables remain identical to those in Table 2. In all columns, we include a prior fixed effect. The t-statistics (in parentheses) are based on standard errors clustered by user and day. We multiply each coefficient by 100.

	Sentiment Change				
	(1)	(2)	(3)	(4)	(5)
Indirect Social Sentiment	18.783***	7.375***	3.078*	3.458*	6.740***
	(13.67)	(5.18)	(1.87)	(1.84)	(2.86)
RavenPack News Sentiment				-0.489	0.731
between Post[0] and Post[1]				(-0.34)	(0.41)
RavenPack News Sentiment				1.533	1.498
24 hours before Post[0]				(0.89)	(1.03)
RavenPack News Sentiment				-0.326	0.097
48 hours before Post[0]				(-0.17)	(0.07)
Bitcoin Return				25.853***	30.882***
				(3.87)	(3.82)
Bitcoin Volatility				-1.577	5.655
				(-0.08)	(0.26)
Forum Sentiment				-2.686*	1.264
				(-1.77)	(0.68)
Prior FE	YES	YES	YES	YES	YES
User FE	NO	YES	YES	YES	NO
Day FE	NO	NO	YES	YES	NO
User X Week FE	NO	NO	NO	NO	YES
Controls	YES	YES	YES	YES	YES
Adjusted R-Squared	0.342	0.348	0.347	0.346	0.327
N	97,724	93,419	93,164	75,548	55,715

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Table 4: Placebo Test

The table displays the results of a placebo test examining the contagion effect. The dependent variable is the sentiment change defined as the revision in sentiment between a user's two consecutive posts post[0] and post[1]. The independent variable is the average sentiment level in a random conversation that the user did not participate in, which occurred between the timestamps of post[0] and post[1]. The control variables remain identical to those in Table 2. In all columns, we include a prior fixed effect. The t-statistics (in parentheses) are based on standard errors clustered by user and day. We multiply each coefficient by 100.

	Sentiment Change				
	(1)	(2)	(3)	(4)	(5)
Random Conversation	2.682***	1.075***	-0.139	0.398	0.256
	(10.41)	(4.11)	(-0.51)	(1.20)	(0.64)
RavenPack News Sentiment				0.212	1.637*
between Post[0] and Post[1]				(0.26)	(1.89)
RavenPack News Sentiment				1.129	1.831**
24 hours before Post[0]				(1.27)	(2.20)
RavenPack News Sentiment				-1.254	0.076
48 hours before Post[0]				(-1.32)	(0.10)
Bitcoin Return				30.482***	34.804***
				(6.31)	(6.35)
Bitcoin Volatility				8.389	-1.761
				(0.61)	(-0.12)
Forum Sentiment				-4.971***	0.420
				(-5.24)	(0.38)
Prior FE	YES	YES	YES	YES	YES
User FE	NO	YES	YES	YES	NO
Day FE	NO	NO	YES	YES	NO
User X Week FE	NO	NO	NO	NO	YES
Controls	NO	NO	NO	YES	YES
Adjusted R-Squared	0.336	0.344	0.345	0.345	0.321
N	257,763	252,098	252,077	198,034	164,058

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Table 5: Effect of Social Sentiment: Heterogeneity across Users

In this table, we explore different users' responses to social sentiment. The dependent variable is the sentiment change defined as the revision in sentiment between a user's two consecutive posts post[0] and post[1]. In columns (1) and (2), we examine whether naive users, indicated by a dummy variable that equals one if the user ranks low on Bitcointalk forum, are more susceptible to social sentiment. In columns (3) and (4), we investigate whether users who have participated in more conversations are less influenced by social sentiment. The user feature measures a user's centrality based on past data on the social network, where two users are considered connected if they have participated in the same thread. In columns (5) and (6), we study whether users who are less informed, defined as those whose post sentiment has a low correlation with future Bitcoin returns in 2 days (correlation $<$ -0.2), are more vulnerable to social sentiment. In columns (1), (3), and (5), we include user and day fixed effects. In columns (2), (4), and (6), we include user-week interactive fixed effects. In all columns, we include a prior fixed effect. The coefficients are multiplied by 100, and the t-statistics (in parentheses) are based on standard errors clustered by user and day.

	Naive Users		Central Users		Less Informed Users	
	(1)	(2)	(3)	(4)	(5)	(6)
User Feature * Social Sentiment	3.585***	3.321***	-0.047***	-0.047***	5.137**	7.378**
	(3.46)	(2.64)	(-3.82)	(-3.41)	(2.24)	(2.44)
Social Sentiment	1.125	2.477***	4.086***	5.466***	2.304***	3.998***
	(1.52)	(2.89)	(7.28)	(7.94)	(4.10)	(5.97)
User Feature			-0.006	0.128	-1.576	-2.069
			(-0.66)	(0.91)	(-1.28)	(-0.76)
Adjusted R-Squared	0.345	0.322	0.345	0.322	0.343	0.323
N	175,750	143,092	175,750	143,092	139,873	118,433
Prior FE	YES	YES	YES	YES	YES	YES
User FE	YES	NO	YES	NO	YES	NO
Day FE	YES	NO	YES	NO	YES	NO
User X Week FE	NO	YES	NO	YES	NO	YES
Controls	YES	YES	YES	YES	YES	YES

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Table 6: Time-varying Sentiment Contagion

In this table, we examine sentiment contagion intensity variations across episodes. The dependent variable is the sentiment change defined as the revision in sentiment between a user's two consecutive posts post[0] and post[1]. In columns (1) and (2), we investigate the impact of informative days on sentiment contagion by defining informative days based on the number of news arrivals documented in the RavenPack database. If there are no news arrivals on a specific day, we assign a value of 0. We construct a dummy variable called *informative days* that takes the value of 1 on a given day if the total number of news arrivals is higher than the sample median. In columns (3) and (4), we investigate how information uncertainty affects sentiment contagion. To identify days of high information uncertainty, we use the standard deviation of news sentiment within the day documented in the RavenPack database. We construct a dummy variable called *high information uncertainty*, which takes the value of 1 if this standard deviation is higher than the sample median and 0 otherwise. If there are no news arrivals on a specific day, we assign a value of 0 to the dummy. In columns (5) and (6), we examine whether Bitcoin return volatility affects sentiment contagion. To do so, we calculate the standard deviation of hourly Bitcoin return within the past 24-hour window before the first post is published. We construct a dummy variable called *high Bitcoin volatility*, which equals 1 if the standard deviation is higher than the sample median and 0 otherwise. In columns (1), (3), and (5), we include user and day fixed effects. In columns (2), (4), and (6), we include user-week interactive fixed effects. In all columns, we include a prior fixed effect. The coefficients are multiplied by 100, and the t-statistics (in parentheses) are based on standard errors clustered by user and day.

	Informative Days		High Uncertainty		High Bitcoin Volatility	
	(1)	(2)	(3)	(4)	(5)	(6)
Episode Feature * Social Sentiment	0.959	0.858	3.714***	3.851**	2.515***	2.919**
	(0.92)	(0.69)	(3.01)	(2.36)	(2.62)	(2.50)
Episode Feature		0.448		-1.059	-0.667	-1.501*
		(0.50)		(-1.10)	(-0.66)	(-1.96)
Social Sentiment	2.664***	3.874***	2.415***	3.613***	1.957***	2.856***
	(3.76)	(4.74)	(4.23)	(5.42)	(2.93)	(3.59)
Adjusted R-Squared	0.345	0.322	0.345	0.322	0.345	0.322
N	175,750	143,092	175,750	143,092	175,750	143,092
Prior FE	YES	YES	YES	YES	YES	YES
User FE	YES	NO	YES	NO	YES	NO
Day FE	YES	NO	YES	NO	YES	NO
User X Week FE	NO	YES	NO	YES	NO	YES
Controls	YES	YES	YES	YES	YES	YES

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Table 7: Social Sentiment and Future Cumulative Returns

This table presents the panel regression analysis of future cumulative Bitcoin returns on social sentiment. The dependent variable is the cumulative returns in future K hours (K = 6, 24, 48, 72, and 168), starting from the ending hour of post[1]. The key predictor is the conversation-level social sentiment. Control variables are identical to those in Table 2. In all columns, we add user and day fixed effect. We also include users' prior, which is a dummy variable that equals 1 if the sentiment in post[0] is non-positive and 0 otherwise. The t-statistics (in parentheses) are based on standard errors clustered by user and day. We multiply each coefficient by 100.

	(1)	(2)	(3)	(4)	(5)
	6 Hours	24 Hours	48 Hours	72 Hours	168 Hours
Social Sentiment	-0.012	-0.094**	-0.119***	-0.111**	-0.114**
	(-0.38)	(-2.17)	(-2.69)	(-2.31)	(-2.57)
RavenPack News Sentiment	-0.733***	-1.859***	-1.831***	-1.369***	-1.786***
between Post[0] and Post[1]	(-3.43)	(-5.61)	(-6.76)	(-5.42)	(-5.09)
RavenPack News Sentiment	-1.566***	-2.586***	-2.778***	-2.223***	-2.624***
24 hours before Post[0]	(-4.28)	(-5.93)	(-7.26)	(-7.66)	(-6.79)
RavenPack News Sentiment	-0.735***	-1.570***	-1.030***	-0.862***	-1.524***
48 hours before Post[0]	(-2.93)	(-4.10)	(-2.77)	(-2.71)	(-4.67)
Bitcoin Return	-21.330***	-35.052***	-39.628***	-41.422***	-41.485***
	(-13.37)	(-8.78)	(-7.81)	(-9.48)	(-10.99)
Bitcoin Volatility	17.717	15.433	45.070***	41.646***	37.695***
	(1.44)	(0.75)	(5.30)	(5.36)	(3.04)
Forum Sentiment	-0.245***	-0.537***	-0.499***	-0.420***	-0.418***
	(-3.82)	(-4.68)	(-4.51)	(-3.68)	(-3.39)
User FE	YES	YES	YES	YES	YES
Date FE	YES	YES	YES	YES	YES
Control For Prior	YES	YES	YES	YES	YES
Adjusted R-Squared	0.268	0.675	0.824	0.878	0.939
N	175,750	175,750	175,750	175,750	175748

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Table 8: Sentiment Contagion of Posts by Different Types of Users

This table presents how users react to posts with varying levels of information. Specifically, we classify posts by user features and compute feature-based "social sentiments." We examine three user features: (i) informed users whose sentiment has a positive correlation (correlation coefficient is higher than 0.1) with future 1-day cumulative Bitcoin returns; (ii) central users who have participated in more conversations; (iii) legendary users whose ranking on Bitcointalk is labeled as "Legendary". In Panel A, we analyze how posts made by users with different features predict future cumulative 24-hour returns, with a similar setting in Table 7. In Panel B, we investigate how users revise their beliefs after reading social sentiment by users of different features. To account for variations, we include user and day fixed effects in all columns of Panel B. We also control for user priors. The t-statistics (shown in parentheses) are based on standard errors that are clustered by user and day. Additionally, we multiply each coefficient by 100 in all panels.

Panel A: Informativeness (future 24-hour returns)

	Informed		Central		Legendary	
	(1)	(2)	(3)	(4)	(5)	(6)
Social Sentiment	0.007		-0.085**		-0.054	
by Featured Users	(0.08)		(-2.23)		(-1.12)	
Social Sentiment		-0.091**		-0.036		-0.061*
by NonFeatured Users		(-2.27)		(-0.99)		(-1.71)
Adjusted R-Squared	0.678	0.678	0.673	0.695	0.674	0.676
N	61974	112938	159205	119231	138938	161144
User FE	YES	YES	YES	YES	YES	YES
Date FE	YES	YES	YES	YES	YES	YES
Controls	YES	YES	YES	YES	YES	YES

Panel B: Response

	Informed		Central		Legendary	
	(1)	(2)	(3)	(4)	(5)	(6)
Social Sentiment	0.483		2.897***		0.797*	
by Featured Users	(0.72)		(5.55)		(1.65)	
Social Sentiment		1.409***		1.925***		2.410***
by NonFeatured Users		(2.62)		(2.84)		(4.97)
User FE	YES	YES	YES	YES	YES	YES
Date FE	YES	YES	YES	YES	YES	YES
Controls	YES	YES	YES	YES	YES	YES
Adjusted R-Squared	0.341	0.345	0.345	0.346	0.347	0.345
N	61,974	112,938	173,477	43,701	138,938	161,144

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Table 9: Confirmation Bias: Echo Chambers

In this table, we use the Probit model to examine how users with different priors selectively participate in conversations. We focus on consecutive pairs of posts (post[0] and then post[1]) published within a 24-hour period by the same user. Our primary explanatory variable is "positive prior", which takes a value of 1 if the user's prior sentiment (i.e., sentiment in post[0]) is positive and 0 otherwise. The dependent variable reflects the attitude of the subsequent conversation in which the user takes part (i.e., the conversation to which post[1] belongs) and is represented by a dummy variable that equals 1 if it signals positive sentiment and 0 otherwise. To determine that the subsequent conversation signals positive sentiment, we employ two methods. In columns (1) and (2), we set the dependent variable to 1 if the thread to which post[1] belongs has a starting post with positive sentiment. In columns (3) and (4), we set the dependent variable to 1 if the social sentiment is positive. The control variables are identical to those in Table 2. We report the average marginal effects. The t-statistics (in parentheses) are computed using standard errors clustered by user and day. Additionally, we multiply each coefficient by 100.

	Positive Sentiment of First Post		Positive Social Sentiment	
	(1)	(2)	(3)	(4)
Positive Prior	6.734***	5.721***	4.251***	3.530***
	(19.64)	(15.97)	(17.84)	(14.37)
Controls	No	Yes	No	Yes
Adjusted R-Squared	0.004	0.019	0.002	0.013
N	190,082	131,407	142,871	123,084

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Table 10: Confirmation Bias: Selective Interpretation

This table investigates whether users' response to social sentiment varies depending on whether the sentiment aligns with their priors. To do so, we define two separate social sentiment variables. Social sentiment (+) takes the positive part and 0 otherwise. Social sentiment (-) is defined in the opposite way. In the first two columns, we analyze the subset of users with positive priors and observe how they respond to positive and negative social sentiment. In column (1), the explanatory variable is the average sentiment in conversations. In column (2), the explanatory variables are two separate sentiment variables. In the last two columns, we investigate the response of users with negative priors to positive and negative social sentiment. In column (3), the explanatory variable is the average sentiment in conversations. In column (4), the explanatory variables are two separate sentiment variables. We control for user priors and include user and day fixed effects in all columns. We report t-statistics in parentheses based on standard errors clustered by user and day. Finally, we multiply each coefficient by 100.

	Positive Priors		NonPositive Priors	
	(1)	(2)	(3)	(4)
Social Sentiment	4.293***		3.307***	
	(6.08)		(4.70)	
Social Sentiment(+)		5.006***		2.803***
		(4.96)		(2.77)
Social Sentiment(-)		2.742*		4.250***
		(1.65)		(2.75)
User FE	YES	YES	YES	YES
Date FE	YES	YES	YES	YES
Controls	YES	YES	YES	YES
Adjusted R-Squared	0.044	0.044	0.248	0.248
N	85,951	85,951	86,230	86,230

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Table 11: Real Effect: Sentiment Contagion and Individual Trading

This table presents the regression analysis of individuals' trading decisions on the sentiment change between Post[0] and Post[1]. In columns (1) and (2), the dependent variable is an indicator for whether there was any trading between a given window. In columns (3), (4) and (5), the dependent variable is an indicator of whether the individual actively increased her portfolio by at least 0.1 Bitcoins. The main explanatory variable is the social sentiment, defined as the average sentiment of others' posts within the conversation between post[0] and post[1]. Columns (3), (4), and (5) condition on windows with active trades. We control for simultaneous events between post[0] and post[1], such as Bitcoin return and volatility, activities in other conversations within the forum, and news arrivals documented in the RavenPack database. We also include lagged news arrivals to capture a delay in response. In all columns, we include user and date fixed effects. The t-statistics (in parentheses) are based on standard errors clustered by user and day. We multiply each coefficient by 100.

	Unconditional		Conditional on trading occurs		
	(1) Probability trade in [t0, t1]	(2) Probability trade in [t1, t1+7days]	(3) Probability buy in [t0, t1]	(4) Probability buy in [t0, t1],t1-t0>2	(5) Probability buy in in [t1, t1+7days]
|Sentiment Change|	0.121	-0.039			
	(1.38)	(-0.22)			
Sentiment Change			7.011*	9.244**	0.576**
			(1.85)	(2.29)	(2.00)
User FE	YES	YES	YES	YES	YES
Date FE	YES	YES	YES	YES	YES
Controls	YES	YES	YES	YES	YES
Adjusted R-Squared	0.086	0.629	0.252	0.261	0.632
N	55,297	55,297	203	189	9,284

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Table 12: Sentiment Contagion Intensity and Market Outcomes

This table presents the predictive power of sentiment contagion intensity (SCI) index on future market outcomes. The main explanatory variable is the SCI indicator, which measures the daily number of users who adjust their sentiment to align with the social sentiment in conversations. We use its logarithmic form and eliminate time-fixed effects by regressing on month and weekday dummies. Polish the sentences: In columns (1) and (2), the primary dependent variable is the future trading volume of Bitcoin (in millions) for the upcoming 1 day and 7 days. We normalize the trading volume by calculating the average of the past two weeks. For columns (3) and (4), the main dependent variable is the future volatility of Bitcoin returns over the next 1 day and 7 days. To account for news arrivals, we include the average levels of sentiment in RavenPack news over the past 14 days as control variables. Additionally, we incorporate two Bitcoin market variables: Bitcoin Volatility and the Number of Transactions. Each variable represents the cumulative sum over the past 14 days. To gauge sentiment in online forums, we calculate the average sentiment of posts on the Bitcointalk forum published in the previous 14 days.The t-statistics (in parentheses) are based on Newey-West corrected standard errors using a lag of 14 days. *, **, and *** represent significance at the 10%, 5%, and 1% levels, respectively. We multiply each coefficient by 100.

	Abnormal Trading Volume		Return Volatility	
	(1)	(2)	(3)	(4)
	1 Day	7 Days	1 Day	7 Days
SCI	401.246***	304.257***	0.536***	0.385***
	(3.26)	(2.69)	(5.55)	(4.62)
Bitcoin Volatility	-2.1e+04***	-2.5e+04***	53.694***	39.923***
in Past 14 Days	(-2.74)	(-2.93)	(7.23)	(8.49)
RavenPack News Sentiment	1396.148**	2002.779***	-0.826**	-1.213**
in Past 14 Days	(2.49)	(3.15)	(-1.96)	(-2.38)
Forum average sentiment	473.546	1075.424	-4.411***	-4.517***
in Past 14 Days	(0.42)	(0.80)	(-3.93)	(-3.54)
Bitcoin Return	2420.147**	2949.043***	3.005***	4.682***
in Past 14 Days	(2.51)	(2.87)	(2.97)	(3.25)
Controls	YES	YES	YES	YES
Adjusted R-Squared	0.025	0.079	0.262	0.385
N	3,664	3,663	3,664	3,663

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Table 13: Optimism in Conversations and Future Market Crashes

In this table, we study how optimism in conversations predicts the occurrence of crashes using the probit model. The dependent variable is a future crash indicator, which takes on a value of 1 if there is any return crash in the next K hours (K =6, 24, 48, and 72) and 0 otherwise. We classify the event of return crash in a given hour if the future Bitcoin return within the subsequent 24 hours is less than -15%. Our key predictor for the future crash is the Optimism variable, and we define it in two distinct ways. In columns (1),(3) and (5), Optimism in summation is the aggregated sentiment in the current hour, where the aggregated sentiment is simply the summation of sentiment across posts published within an hour. In columns (2), (4) and (6), Optimism in fraction is the fraction of hours with positive aggregated sentiment in the past 7*24 hours, where the aggregated sentiment is simply the summation of sentiment across posts published within an hour. We report the average marginal effects. We report t-statistics in parentheses based on standard errors clustered by day. Finally, we multiply each coefficient by 100.

	(1)	(2)	(3)	(4)	(5)	(6)
	6 Hours	6 Hours	24 Hours	24 Hours	72 Hours	72 Hours
Optimism in Summation	0.179*		0.419**		1.312***	
	(1.95)		(2.16)		(3.58)	
Optimism in Fraction		0.414**		1.182***		2.782***
		(2.24)		(3.24)		(4.38)
RavenPack News Sentiment	0.039	-0.184	0.519	-0.215	1.002	-0.751
in Past 7*24 Hours	(0.06)	(-0.29)	(0.44)	(-0.19)	(0.50)	(-0.39)
Bitcoin Return Volatility	170.494***	167.886***	441.053***	422.925***	1021.164***	981.951***
in Past 7*24 Hours	(4.78)	(4.76)	(6.16)	(6.27)	(8.16)	(8.18)
Cumulative Bitcoin Return	-4.347	-3.308	-12.444*	-9.780	-18.378	-10.586
in Past 7*24 Hours	(-1.02)	(-0.78)	(-1.65)	(-1.27)	(-1.44)	(-0.82)
Adjusted R-Squared	0.109	0.116	0.127	0.141	0.140	0.153
N	17953	17953	17953	17953	17953	17953

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Table 14: Optimism in Conversations and Future Volatility

In this table, we study how optimism in conversations predicts future number of transactions volume. The dependent variable volatility is the standard deviation of hourly returns in the next K hours (K = 6, 24, 72). We adjust the standard deviation to the daily level. Our key predictor for the future crash is the Optimism variable, and we define it in two distinct ways. In columns (1), (3) and (5), Optimism in summation is the aggregated sentiment in the current hour, where the aggregated sentiment is simply the summation of sentiment across posts published within an hour. In columns (2), (4) and (6), Optimism in fraction is the fraction of hours with positive aggregated sentiment in the past 7*24 hours, where the aggregated sentiment is simply the summation of sentiment across posts published within an hour. We multiply each coefficient by 100. In this Table, we use Bootstrap standard errors from a simulation of 1000 times.

	(1)	(2)	(3)	(4)	(5)	(6)
	6 Hours	6 Hours	24 Hours	24 Hours	72 Hours	72 Hours
Optimism in Summation	0.288***		0.260***		0.221***	
	(9.29)		(10.37)		(10.83)	
Optimism in Fraction		0.207***		0.215***		0.203***
		(6.75)		(9.01)		(10.44)
RavenPack News Sentiment	-0.351***	-0.442***	-0.121	-0.232**	0.137	0.023
in Past 7*24 Hours	(-2.96)	(-3.52)	(-1.12)	(-2.02)	(1.55)	(0.25)
Bitcoin Return Volatility	426.657***	433.016***	405.186***	409.712***	363.787***	366.767***
in Past 7*24 Hours	(28.08)	(28.43)	(29.94)	(30.06)	(32.86)	(32.71)
Cumulative Bitcoin Return	-17.934**	-16.988**	-15.784***	-14.915***	-8.160**	-7.411**
in Past 7*24 Hours	(-2.57)	(-2.43)	(-3.26)	(-3.07)	(-2.39)	(-2.17)
Adjusted R-Squared	0.286	0.282	0.328	0.324	0.331	0.328
N	17,953	17,953	17,953	17,953	17,953	17,953

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Table 15: Summary Statistics: Bubble vs Non-Bubble Episodes

This table tabulates the summary statistics of the key features in bubble and non-bubble episodes, and the difference of the key variables in bubble and non-bubble episodes. To identify bubbles, we identify days when the price of Bitcoin experiences a significant and rapid increase, leading to local peaks. Our approach focuses on selecting days where the cumulative Bitcoin return over the past quarter exceeds 200%. Once we have identified a series of consecutive days meeting these criteria, we group them together to prepare for a bubble episode. Within each group, we designate the day with the highest Bitcoin price as the peak day and identify the day occurring two quarters prior to the peak day as the start date. The bubble episode encompasses the time period between the start date and the peak date. In Panel A, we describe market variables. In Panel B, we describe social interactions. In Panel C, we describe sentiment contagions. We rely on seemingly unrelated regression (SUR) to test the joint significance of differences in all features. We report the joint F-statistic and its corresponding p-value. Our sample spans from Jan 5, 2012 to July 30, 2022.

	Bubble Episode		Non-Bubble Episode		Bubble Episode minus Non-Bubble Episode	
Features	Mean	Std	Mean	Std	Difference	t-statistic
Panel A: Market Variables						
Daily Return(Annualized)	4.050	17.66	0.491	15.72	3.56	5.352
Return Volatility(Within Day)	0.040	0.03	0.037	0.04	0.003	2.413
Total Dollar Volume (in millions)	94.071	148.80	47.503	77.63	46.568	11.784
RavenPack News Sentiment	0.168	0.42	0.040	0.43	0.127	5.894
Abnormal Google Search Volume	0.085	0.48	0.017	0.40	0.068	3.961
Panel B: Social Interactions						
Average Sentiment	0.305	0.11	0.272	0.12	0.033	6.761
Standard Deviation of Sentiment	0.657	0.05	0.672	0.05	-0.015	-6.635
Number of Posts	188.764	147.95	175.713	163.66	13.052	1.968
Number of Users	133.504	105.61	116.261	114.97	17.243	3.692
Total Number of Positive Posts	104.671	87.03	90.834	87.47	13.837	3.837
Fraction of Posts with Positive Sentiment	0.543	0.09	0.516	0.09	0.028	7.563
Fraction of Sophisticated Users	0.301	0.10	0.359	0.14	-0.058	-10.523
Panel C: Sentiment Contagion						
Number of Positively Infected Users	16.704	16.24	15.335	17.33	1.369	1.938
Number of Negatively Infected Users	3.203	4.63	3.813	5.84	-0.61	-2.629
Joint F-statistic						15.49
p-value (Probability>F)						0.00

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Table A1: Contagion Effect: Ex-Post Sentiment as Dependent Variable

The dependent variable, ex-post sentiment, refers to the sentiment in the first post of two consecutive posts (post[0] and post[1]). The main explanatory variable is social sentiment, which is the average sentiment of others' posts within the conversation published between the timestamps of post[0] and post[1]. We control for simultaneous events between post[0] and post[1], such as Bitcoin return and volatility, activities in other conversations within the forum, and news arrivals documented in the RavenPack database. We also include lagged news arrivals to capture a delay in response. In all columns, we include a prior fixed effect. The t-statistics (in parentheses) are based on standard errors clustered by investor and day. We multiply each coefficient by 100.

	Ex-Post Sentiment					
	(1)	(2)	(3)	(4)	(5)	(6)
Social Sentiment	11.036***	6.671***	3.485***	3.948***	5.243***	4.681***
	(26.20)	(17.76)	(9.06)	(8.20)	(9.43)	(6.81)
Prior Sentiment		0.616***	-0.491**	-0.543**	-11.369***	-25.756***
		(2.68)	(-2.17)	(-2.16)	(-25.66)	(-37.98)
RavenPack News Sentiment				0.506	2.236***	0.712
between Post[0] and Post[1]				(0.65)	(2.71)	(0.60)
RavenPack News Sentiment				1.546*	2.287***	0.574
24 hours before Post[0]				(1.70)	(2.84)	(0.38)
RavenPack News Sentiment				-0.947	0.182	-1.491
48 hours before Post[0]				(-1.04)	(0.23)	(-1.06)
Bitcoin Return				31.323***	36.208***	30.815***
				(6.66)	(7.01)	(5.52)
Bitcoin Volatility				-4.750	-8.252	-22.528
				(-0.29)	(-0.49)	(-1.17)
Forum Sentiment				-5.887***	1.780*	1.212
				(-6.61)	(1.93)	(1.07)
User FE	NO	YES	YES	YES	YES	YES
Day FE	NO	NO	YES	YES	YES	YES
User X Week FE	NO	NO	NO	NO	YES	NO
User X Day FE	NO	NO	NO	NO	NO	YES
Controls	NO	NO	NO	YES	YES	YES
Adjusted R-Squared	0.005	0.033	0.038	0.039	0.050	0.108
N	218,270	212,647	212,623	175,750	143,092	98,910

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Table A2: Contagion Effect: Max of Sentiment

In this table, we use the max of sentence sentiment to measure sentiment at the post level. We present the panel regression analysis of sentiment change on social sentiment. The dependent variable is the sentiment change defined as the revision in sentiment between a user's two consecutive posts post[0] and post[1]. The main explanatory variable is the social sentiment defined as the average sentiment of others' posts within the conversation, published between the timestamps of post[0] and post[1]. We control for simultaneous events between post[0] and post[1], such as Bitcoin return and volatility, activities in other conversations within the forum, and news arrivals documented in the RavenPack database. We also include lagged news arrivals to capture a delay in response. In all columns, we include a prior fixed effect. The t-statistics (in parentheses) are based on standard errors clustered by user and day. We multiply each coefficient by 100.

	Sentiment Change					
	(1)	(2)	(3)	(4)	(5)	(6)
Social Sentiment	9.164***	5.265***	2.562***	2.726***	4.050***	3.405***
	(22.65)	(13.63)	(6.34)	(5.50)	(6.85)	(4.48)
RavenPack News Sentiment				0.463	2.338**	2.029
between Post[0] and Post[1]				(0.49)	(2.27)	(1.26)
RavenPack News Sentiment				1.179	1.660	-0.567
24 hours before Post[0]				(1.08)	(1.62)	(-0.28)
RavenPack News Sentiment				-1.758	0.290	-2.585
48 hours before Post[0]				(-1.61)	(0.32)	(-1.37)
Bitcoin Return				37.514***	41.362***	41.480***
				(6.93)	(6.68)	(6.28)
Bitcoin Volatility				9.714	6.044	-18.674
				(0.57)	(0.30)	(-0.74)
Forum Sentiment				-5.321***	1.330	0.444
				(-5.37)	(1.15)	(0.31)
Adjusted R-Squared	0.376	0.385	0.385	0.386	0.372	0.367
N	218,268	212,647	212,623	175,750	143,092	98,910
User FE	NO	YES	YES	YES	NO	NO
Day FE	NO	NO	YES	YES	NO	NO
Prior FE	YES	YES	YES	YES	YES	YES
User X Week FE	NO	NO	NO	NO	YES	YES
Controls	NO	NO	NO	YES	YES	YES

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Table A3: Contagion Effect: Sum of Sentiment

In this table, we use the summation of sentence sentiment to measure sentiment at the post level. We present the panel regression analysis of sentiment change on social sentiment. The dependent variable is the sentiment change defined as the revision in sentiment between a user's two consecutive posts post[0] and post[1]. The main explanatory variable is the social sentiment defined as the average sentiment of others' posts within the conversation, published between the timestamps of post[0] and post[1]. We control for simultaneous events between post[0] and post[1], such as Bitcoin return and volatility, activities in other conversations within the forum, and news arrivals documented in the RavenPack database. We also include lagged news arrivals to capture a delay in response. In all columns, we include a prior fixed effect. The t-statistics (in parentheses) are based on standard errors clustered by user and day. We multiply each coefficient by 100.

	Sentiment Change					
	(1)	(2)	(3)	(4)	(5)	(6)
Social Sentiment	8.978***	5.091***	2.550***	2.788***	3.811***	2.268**
	(20.08)	(11.19)	(5.42)	(4.93)	(5.49)	(2.37)
RavenPack News Sentiment				0.927	3.516**	3.345
between Post[0] and Post[1]				(0.62)	(2.26)	(1.30)
RavenPack News Sentiment				0.379	1.953	-3.051
24 hours before Post[0]				(0.23)	(1.28)	(-1.00)
RavenPack News Sentiment				-3.092**	-0.551	-3.612
48 hours before Post[0]				(-2.03)	(-0.43)	(-1.31)
Bitcoin Return				47.551***	52.880***	37.394*
				(4.35)	(5.22)	(1.83)
Bitcoin Volatility				17.104	10.720	-15.418
				(0.80)	(0.36)	(-0.46)
Forum Sentiment				-5.194***	1.171	-0.075
				(-4.75)	(0.97)	(-0.05)
Adjusted R-Squared	0.306	0.310	0.310	0.311	0.281	0.250
N	218,268	212,647	212,623	175,750	143,092	98,910
User FE	NO	YES	YES	YES	NO	NO
Day FE	NO	NO	YES	YES	NO	NO
Prior FE	YES	YES	YES	YES	YES	YES
User X Week FE	NO	NO	NO	NO	YES	YES
Controls	NO	NO	NO	YES	YES	YES

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Table A4: Contagion Effect: Same Conversation

This table presents the panel regression analysis of sentiment change on social sentiment when post[0] and post[1] are published in the same thread within a 24-hours window. The dependent variable is the sentiment change defined as the revision in sentiment between a user's two consecutive posts post[0] and post[1]. The main explanatory variable is the social sentiment defined as the average sentiment of others' posts within the conversation, published between the timestamps of post[0] and post[1]. We control for simultaneous events between post[0] and post[1], such as Bitcoin return and volatility, activities in other conversations within the forum, and news arrivals documented in the RavenPack database. We also include lagged news arrivals to capture a delay in response. In all columns, we include a prior fixed effect. The t-statistics (in parentheses) are based on standard errors clustered by user and day. We multiply each coefficient by 100.

	Sentiment Change				
	(1)	(2)	(3)	(4)	(5)
Social Sentiment	5.461***	4.225***	2.165***	1.821*	3.652***
	(8.91)	(6.49)	(3.14)	(1.91)	(3.03)
RavenPack News Sentiment				-1.657	0.115
between Post[0] and Post[1]				(-0.81)	(0.05)
RavenPack News Sentiment				0.662	-1.265
24 hours before Post[0]				(0.33)	(-0.63)
RavenPack News Sentiment				0.184	-0.037
48 hours before Post[0]				(0.09)	(-0.02)
Bitcoin Return				33.676***	35.570***
				(3.81)	(3.54)
Bitcoin Volatility				20.998	-17.294
				(0.74)	(-0.51)
Forum Sentiment				-2.373	2.054
				(-1.52)	(1.20)
Adjusted R-Squared	0.341	0.340	0.335	0.335	0.323
N	73,466	70,814	70,388	53,283	42,902
User FE	NO	YES	YES	YES	NO
Day FE	NO	NO	YES	YES	NO
Prior FE	YES	YES	YES	YES	YES
User X Week FE	NO	NO	NO	NO	YES
Controls	NO	NO	NO	YES	YES

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Table A5: Contagion Effect: Sentiment Change over the Next 12, 48, and 72 Hours

This table presents panel regression analysis of sentiment change on social sentiment when post[0] and post[1] are published within a 12-, 48- and 72-hours window. The dependent variable is the sentiment change defined as the revision in sentiment between a user's two consecutive posts post[0] and post[1]. The main explanatory variable is the social sentiment defined as the average sentiment of others' posts within the conversation, published between the timestamps of post[0] and post[1]. We control for simultaneous events between post[0] and post[1], such as Bitcoin return and volatility, activities in other conversations within the forum, and news arrivals documented in the RavenPack database. We also include lagged news arrivals to capture a delay in response. In all columns, we include a prior fixed effect. The t-statistics (in parentheses) are based on standard errors clustered by user and day. We multiply each coefficient by 100.

	12 Hours		48 Hours		72 Hours	
	(1)	(2)	(3)	(4)	(5)	(6)
Social Sentiment	2.575***	3.936***	3.975***	5.024***	4.235***	5.158***
	(4.31)	(5.19)	(8.45)	(9.00)	(9.17)	(9.70)
RavenPack News Sentiment	0.627	2.678*	0.220	2.470***	0.192	2.440***
between Post[0] and Post[1]	(0.56)	(1.88)	(0.29)	(3.32)	(0.28)	(3.57)
RavenPack News Sentiment	1.211	1.316	0.183	1.433**	0.475	1.168*
24 hours before Post[0]	(0.90)	(1.10)	(0.26)	(2.15)	(0.80)	(1.94)
RavenPack News Sentiment	-1.692	-0.107	-0.806	-0.110	-0.200	-0.162
48 hours before Post[0]	(-1.11)	(-0.09)	(-1.13)	(-0.17)	(-0.33)	(-0.28)
Bitcoin Return	28.611***	27.871***	24.719***	31.343***	21.591***	28.988***
	(4.56)	(3.67)	(5.32)	(5.94)	(5.37)	(6.29)
Bitcoin Volatility	5.691	9.019	14.377	-2.341	15.457	-1.280
	(0.31)	(0.49)	(0.97)	(-0.13)	(1.06)	(-0.07)
Forum Sentiment	-3.348***	0.895	-5.744***	1.636	-6.421***	1.413
	(-2.99)	(0.67)	(-5.52)	(1.43)	(-6.17)	(1.25)
Adjusted R-Squared	0.345	0.326	0.348	0.321	0.350	0.319
N	108,372	84,010	245,944	206,206	283,588	239,714
User FE	YES	NO	YES	NO	YES	NO
Day FE	YES	NO	YES	NO	YES	NO
Prior FE	YES	YES	YES	YES	YES	YES
User X Week FE	NO	YES	NO	YES	NO	YES
Controls	YES	YES	YES	YES	YES	YES

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Table A6: Post Decisions and Recent Bitcoin Returns

This table presents the association between users' post decisions and Bitcoin returns in the past 14 days. In column (1) and (2), the dependent variable $PostDecisions_{i,t}$ is a dummy variable that equals one if user i makes at least one post on day t and zero otherwise. We focus on users' active days after their registration. In column (3) to (6), we investigate how, conditional on users' post decisions, recent Bitcoin returns affect sentiment in posts . In column (3) and (4), the dependent variable $PositivePosts_{i,t}$ equals to one if user i publishes a positive posts on day t and zero otherwise. In column (5) and (6), the dependent variable $PessimisticPosts_{i,t}$ equals to one if user i publishes a negative posts on day t and zero otherwise. We calculate the average levels of sentiment in RavenPack news in the past 14 days as controls for news arrivals. We control for market information by controlling for two Bitcoin market variables: Bitcoin Volatility and Number of Transactions. Each of two variable is the cumulative sum in the past 14 days. We calculate forum average sentiment as the average sentiment of posts on the Bitcointalk forum published in the past 14 days. In all columns, we add date fixed effect. We multiply each coefficient by 100. The t-statistics (in parentheses) are based on standard errors clustered by user and date. *, **, and *** represent significance at the 10%, 5%, and 1% levels, respectively.

	(1) Post Decisions	(2) Post Decisions	(3) Positive Posts	(4) Positive Posts	(5) Pessimistic Posts	(6) Pessimistic Posts
Bitcoin Return	0.210***	0.299***	6.677***	2.549***	-4.920***	-2.373***
in Past 14 Days	(3.96)	(4.92)	(10.01)	(3.81)	(-9.54)	(-4.59)
Bitcoin Volatility		1.016***		2.728**		0.047
in Past 14 Days		(11.12)		(2.28)		(0.05)
RavenPack News Sentiment		-0.042***		2.446***		-1.986***
in Past 14 Days		(-3.15)		(10.33)		(-11.86)
Forum average sentiment		-0.231**		43.072***		-26.911***
in Past 14 Days		(-2.08)		(19.20)		(-15.59)
User FE	YES	YES	YES	YES	YES	YES
Date FE	NO	NO	NO	NO	NO	NO
Controls	YES	YES	YES	YES	YES	YES
Adjusted R-Squared	0.055	0.055	0.039	0.039	0.020	0.023
N	122,046,111	121,971,272	649,710	403,816	649,710	403,816

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Table A7: Fraction of Positive and Negative Posts and Recent Bitcoin Returns

This table presents the association between the fraction of positive and negative posts within a day and Bitcoin returns in the past 14 days. In column (1) and (2), the dependent variable is the fraction of positive posts within one day. In column (3) and (4), the dependent variable is the fraction of negative posts within one day. We calculate the average levels of sentiment in RavenPack news in the past 14 days as controls for news arrivals. We control for Bitcoin Volatility which is based on the returns in the past 14 days. We multiply each coefficient by 100. The t-statistics (in parentheses) are based on standard errors clustered by date. *, **, and *** represent significance at the 10%, 5%, and 1% levels, respectively.

	(1)	(2)	(3)	(4)
Bitcoin Return in Past 14 Days	5.435***	4.383***	-4.547***	-3.804***
	(6.04)	(5.41)	(-7.74)	(-7.28)
Bitcoin Volatility in Past 14 Days		-9.330***		6.631***
		(-8.17)		(9.02)
RavenPack News Sentiment in Past 14 Days		2.448***		-1.731***
		(7.56)		(-8.13)
Adjusted R-Squared	0.014	0.042	0.025	0.060
N	3,742	3,742	3,742	3,742

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